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Exploring complexity matching and asynchrony dynamics in synchronized and syncopated task performances.

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Abstract

When two people synchronize their rhythmic behaviors (e.g., finger tapping; walking) they match one another not only at a local scale of beat-to-beat intervals, but also at a global scale of the complex (fractal) patterns of variation in their interval series. This “complexity matching” had been demonstrated in a variety of timing behaviors, but the current study was designed to address two important gaps in previous research. First, very little was known about complexity matching outside of synchronization tasks. This was important because different modes are associated with differences in the strength of coordination and the fractal scaling of the task performance. Second, very little was known about the dynamics of the asynchrony series. This was important because asynchrony is a variable directly quantifying the coordination between the two timing behaviors and the task goal. So, the current study explored complexity matching in both synchronized and syncopated finger tapping tasks, and included analyses of the fractal scaling of the asynchrony series. Participants completed an interpersonal finger tapping task, in both synchronization and syncopation conditions. The magnitude of variation and the exact power law scaling of the tapping intervals were manipulated by having one participant tap in time with a metronome. Complexity matching was most stable when there was sufficient variation in the task behavior and when a persistent scaling dynamic was presented. There were, however, several interesting differences between the two coordination modes, in terms of the heterogeneity of the complexity matching effect and the scaling of the asynchronies. These findings raised a number of important points concerning how to approach and understand the interaction of inherently complex systems.

Keywords

complexity matching; fractal scaling; syncopation; interpersonal coordination; finger tapping

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1. Introduction

Whether the question is how the nervous system regulates behavior within a changing environment, how a child learns to speak from engaging with her parents, or how a stigmatized person addresses his difficulties within society, human behavior inherently involves the interaction of complex, dynamical systems. These interactions involve patterns of co-regulation and co-evolution, making it difficult to understand these systems outside the context of their interactions with one another. For instance, the child's pattern of speaking changes with linguistic feedback from her parents, but her parents' patterns of feedback also change according to the patterns in her developing speech (Kavanaugh & Jirkovsky, 1982). So, there is a need to better understand the basic principles underlying these kind of multiscale, interdependent relationships between systems. One avenue of research that offers some insight into this issue involves how interacting people tend to match one another's complex patterns of variation in task performance.

Research has demonstrated that many different kinds of human performances display fractal scaling (Bassingthwaighte, Liebovitch, & West, 1994; Diniz et al., 2011; Van Orden, Kloos, & Wallot, 2011). That is, the pattern of variation across many performances of a task is not random, but instead obeys a power-law scaling relationship, such that small fluctuations in performance show the same basic pattern as large fluctuations (Eke et al., 2000; Holden, 2005). For instance, when someone rhythmically taps their finger, some taps fall below the average tapping interval and some fall above. Importantly, the *pattern of deviations* shows coherent trends in which the participant speeds up for a short while, and then slows down again. More importantly, these trends obey a fractal pattern, sometimes called “ $1/f$ ” or “pink” noise, where the relationship between the size of the deviation and how many taps are involved in that trend is the same across all scales. These kind of fractal scaling relationships are characteristic of natural systems (Mandelbrot, 1983; West & Deering, 1995), and many studies have shown fractal scaling in a variety of human physiological and behavioral signals (e.g., Linkenkaer-Hansen, Nikouline, Palva, & Ilmonniemi, 2001; Schenkel, Zhang, & Zhang, 1993; Van Orden, Holden, & Turvey, 2003; 2005). Although there is debate concerning the origin and meaning of fractal scaling (see Diniz et al., 2011; Van Orden et al., 2011), finding pink noise in a behavior has generally become accepted as a signature of system complexity (Diniz et al., 2011).

More recent research has shown that the degree of fractal scaling in a behavior also tends to be closely matched to the scaling in the behavior of coupled systems (Delignières & Marmelat, 2014; Fine, Likens, Amazeen, & Amazeen, 2015; Hennig, 2014; Stephen, Stepp, Dixon, & Turvey, 2008). This “complexity matching” is typically assessed by correlating the fractal scaling exponents between interacting systems. That is, the degree of fractal scaling can be quantified by certain time series analyses, which yield a scaling exponent (α). A time series of task performances (e.g., inter-tap intervals) are said to show fractal scaling when the scaling exponent is near $\alpha = 1$, and are said to show a random pattern when the exponent is near $\alpha = 0$. So, the bivariate correlation of scaling exponents can show that the degree of fractal scaling of one system is dependent on that of another, with greater correlations indicating stronger complexity matching between the interacting systems. Complexity matching has been observed in several kinds of simplified laboratory tasks, such as

participants tapping their fingers in time with chaotic metronomes (Stephen et al., 2008) or two participants swinging pendulums in time with one another (Marmelat & Delignières, 2012). Importantly, research also suggests that similar phenomena are present in more naturalistic and complicated forms of behavior. This includes rhythmic behaviors similar to the laboratory tasks (e.g., team rowing; Den Hartigh, Marmelat, & Cox, 2018) and non-rhythmic behaviors, such as the actions of two partners interacting in a virtual reality task (Zapata-Fonseca, Dotov, Fossion, & Froese, 2016). Complexity matching, and more broadly multiscale coordination, has also been demonstrated in the bodily movements of two people interacting verbally (Schmidt, Nie, Franco, & Richardson, 2014), as well as in the timing (Abney, Paxton, Dale, & Kello, 2014) and syntactic complexity (Xu & Reitter, 2016) of the speech itself. In fact, research supports that complexity matching might play a critical role in the kind of language development processes in the example above (Abney, Warlaumont, Oller, Wallot, & Kello, 2017).

One of the central questions in this field of research regards the nature of coordination underlying the complexity matching effect. Paralleling the debates about fractal scaling in human behavior more broadly (Diniz et al., 2011), some researchers proposed that complexity matching reflects a more general property of system interdependence, while others argued for localizable, mechanistic dynamics. Several of early studies supported that complexity matching resulted from a “global” coordinative process (e.g., Marmelat & Delignières, 2012; Stephen et al., 2008). These studies found that although there was a strong correlation between the scaling exponents defining the two interval time series, there was considerably weaker correlation between the time series themselves, on a beat-to-beat basis. That is, one system’s behavior at any particular time did not strongly predict the other system’s behavior at that time, but the overall structure of one system’s behavior, across all timescales, was strongly predictive of the other system’s overall structure. These findings lead some researchers to consider some “global” coordinative process that operated across many timescales, without producing any strong “local” coordination.

Other researchers have argued that a local, or “short-term”, coordinative process is sufficient to create complexity matching (Hennig, 2014; Konvalinka, Vuust Roepstorff, & Frith, 2010). For instance, Torre, Varlet, and Marmelat (2013) demonstrated that a mechanistic model, based exclusively on a short-term error-correction process, could produce both the strong complexity matching and the relatively weak cross-correlation between series. Similarly, Coey, Washburn, Hassebrock and Richardson (2016) demonstrated that although the cross-correlation between series was relatively weak overall, the strength of the peak cross-correlation was significantly related to the strength of the complexity matching observed.

One way to better understand the coordination involved in complexity matching is to explore experimental manipulations, which are known to affect coordination dynamics generally, but have yet to be implemented in complexity matching research. Notably, the existing literature was almost exclusively focused on synchronization tasks (but see Abney et al., 2014, Coey et al., 2016). This was an important limitation because different modes of coordination can be more or less stable. Ample research has shown that synchronized (or inphase) modes are more stable than syncopated (or antiphase) modes of coordination (see Repp, 2005; Schmidt & Richardson, 2008; for reviews). The difference in stability between these modes can be

demonstrated by increasing the movement frequency. When in the syncopated mode, increasing movement frequency past a critical value produces a spontaneous transition to the synchronized mode, in both continuous and discrete timing tasks (e.g., Schmidt & Turvey, 1994; Volman & Geuze, 2000). The opposite transition, from synchronization to syncopation does not occur spontaneously, even when the frequency is decreased to a point where stable syncopated, antiphase behavior is possible (Haken, Kelso, & Bunz, 1985). So, as a manipulation of the strength of coordination, comparing syncopation to synchronization might help clarify the coordinative dynamics underlying complexity matching. This specific manipulation complements an earlier complex matching study (Marmelat & Delignières, 2012), which manipulated the strength of the coordination within a synchronization task by varying the perceptual coupling between participants. It also complements a recent study (Coey et al., 2016), which demonstrated complexity matching in interpersonal syncopated tapping, but lacked a comparison synchronization condition.

Another important limitation in the complexity matching literature is that previous studies have paid limited or no attention to the temporal patterning and fractal scaling of the asynchrony (or relative phase) signal. That is, these studies have focused on the correlation between the scaling exponents defining the “interval” series (e.g., the inter-tap intervals of the participant and the inter-onset intervals of the metronome). The asynchrony series is also a very important index of *coordination*, because it is mutually defined by the behavior of the two systems. At a minimum, the asynchrony provides an additional piece of information extremely relevant for understanding the coordination involved in complexity matching, as it directly quantifies the coordinative outcome of the task. Other researchers have made similar arguments with respect to examining the fractal scaling of the asynchrony series. Chen, Ding, and Kelso (1997) demonstrated that when participants synchronized with an isochronous metronome, it was the “error” (i.e., asynchrony) series, and not the interval series as in self-paced tapping (e.g., Gilden, 2001), which displayed fractal scaling. They called the asynchrony series the “fundamental time series” for tasks in which participants coordinate their behavior to an external timing signal, adding that this is a helpful concept when “several interrelated dynamical variables all appear to provide an adequate description of the problem” (p. 4503). Moreover, Chen, Ding, and Kelso (2001) found that the fractal scaling of the asynchrony series was influenced by the mode of coordination with the metronome, being higher when participants syncopated than when they synchronized to the metronome, as well as the specific strategy participants used to maintain syncopated coordination. So, the fractal scaling of the asynchrony is sensitive to factors which may, or may not, also influence the strength of complexity matching. Better understanding the relationships between the asynchrony and the interval dynamics would thus help clarify system organization and coordination that produces complexity matching.

The current study was designed to address these limitations and thoroughly explore the emergence and strength of complexity matching outside of synchronization tasks. We replicated and extended the design of an earlier study (Marmelat, Delignières, Torre, Beek, & Daffertshofer, 2014) to also include a syncopation condition. Specifically, we had participants complete an interpersonal tapping task, in which one participants’ behavior (the “leader”) was synchronized to a discrete metronome stimulus, and the other participant (the “follower”) coordinated with the taps of the leader in either a synchronized or syncopated

pattern. We also manipulated the overall level of variation in the metronome (Experiment 1), and the fractal scaling of the metronome inter-onset intervals (Experiment 1 and 2). We examined the effects of these manipulations on (1) the complexity matching between the interval time series of the metronome, leader, and follower systems, (2) the fractal scaling of the asynchrony series between the three systems, and (3) the short-term cross-correlations between the interval series and the asynchrony series. The results of these analyses were used to characterize the coordination involved in complexity matching. For instance, the degree of “local” coordination was examined by associations between the changes in the strength of complexity matching, brought on by the experimental manipulations, and changes in the strength of the peak crosscorrelations. Comparisons of changes in the strength of complexity matching and changes in the fractal scaling of the asynchrony series were also helpful in better understanding the coordination between these complex systems.

2. Experiment 1

Experiment 1 was designed to determine how the overall amount of variability and fractal scaling of a metronome pacing signal influenced complexity matching during both synchronized and syncopated finger tapping. Based on past research (e.g., Coey, Hassebrock, Kloos, & Richardson, 2015), the coefficient of variation in self-paced tapping was estimated to be approximately 6%. The goal of the current study was to explore a wide range of levels of variability in the pacing signal around this general level of variability in the uncoupled, self-paced behavior to assess for differences in the strength of complexity matching. This design was partially based on the past work of Marmelat and colleagues (2014). Their first experiment was also concerned with how variability in the pacing signal affected complexity matching, but they only used a synchronization task and only investigated one level of fractal scaling in the pacing signal. Based on their findings, we hypothesized strong complexity matching would only appear above some minimum amount of variation in the pacing signal, at least within the synchronization task. A previous study (Coey et al., 2016) demonstrated that complexity matching was equally evident in both syncopation and synchronization tapping, but it was unclear whether the metronome variation would influence the coordination modes differently. Additionally, we were interested in assessing the effect of these manipulations on the fractal scaling of the asynchrony between the participants and the pacing signal, as well as the short-term correlation between the paired inter-event intervals and the asynchronies. This information was expected to provide a picture of the short-term (or “local”) interactions between the interrelated dynamical task variables and their relationships to complexity matching.

2.1. Method

2.1.1. Participants—A total of 12 undergraduate and graduate student volunteers from the Department of Psychology participated in the experiment. The study required multiple sessions, so they were assigned to pairs ($N = 6$) according to their respective availabilities. The mean age of the participants was 21.1 years, 50% of the participants were male, and 91.7% of the participants were right-handed.

2.1.2. Apparatus—The participants' tapping was recorded using two USB MIDI keyboards (± 5 ms error) connected to a PC computer running Ableton Live (v. 8.2) software (Ableton, Berlin Germany). This software simultaneously recorded the participants' taps and presented the metronome stimuli to the participants through pairs of headphones.

2.1.3. Design—The metronome stimuli were created to have an average inter-onset interval of 800ms. The magnitude of variation around this average was manipulated within-subjects, with every pair experiencing metronomes with five different coefficients of variation (CoV) of 0.5% ($SD = 4$ ms), 1% ($SD = 8$ ms), 3% ($SD = 24$ ms), 6% ($SD = 48$ ms), and 9% ($SD = 72$ ms). The fractal scaling of the metronomes was also manipulated within-subjects. Every pair experienced both white noise, random ($\alpha \approx 0$) metronomes and pink noise, persistent ($\alpha \approx 1$) metronomes¹. This design resulted in a total of 10 metronome conditions. Three different sets of metronome stimuli were generated to ensure that any effects observed were not artifacts of any particular metronome series, and each participant pair experienced only one of these sets. Finally, the coordination between the participants was manipulated within-subjects, with every pair of participants completing both synchronization and syncopation trials. The trials were completed in blocks. Half of the pairs first completed 10 trials in synchronization, then 10 trials in syncopation. The other half completed the trials in the opposite order. This resulted in a total of 20 trials for every participant pair. The metronome series consisted of 600 beats, and each trial lasted approximately 8 minutes. The full experiment lasted approximately 2 hours and 40 minutes.

2.1.4. Procedure—Following informed consent, the two participants in each pair were randomly assigned to the role of “leader” and “follower”. The participants were seated at a table in front of their respective keyboards, facing each other across the table, wearing noise-cancelling headphones, with a small screen occluding the follower's hands from the leader's vision. The leader was instructed to always tap in synchrony with the metronome they heard through the headphones. The follower was told that they would not hear anything through the headphones, but instead to tap in synchrony or in syncopation with the leader, depending on the coordination mode block. Within each block of 10 trials, the order of presentation of metronomes with different magnitudes of variation and fractal scaling values was randomized. When the pair completed their first block of 10 trials, the mode of coordination between the participant switched, and the follower was instructed to now tap in the other pattern with the leader. The participants were allowed short breaks between trials and completed the total of 20 trials across 3 to 5 sessions depending on time availability.

2.1.5. Data Analysis—The primary research questions involved both the inter-event interval dynamics of the metronome stimulus (inter-onset intervals; IOI) and the participants (inter-tap intervals; ITI), as well as the asynchrony (ASY) dynamics. Preparing the time series required several pre-processing procedures. First, observations were matched across interacting systems. If the systems were processed independently of one another, without

¹The metronome inter-onset interval series were generated using the inverse FFT function in Matlab to generate a candidate time series with a desired scaling exponent. The exact scaling exponent defining the output series can deviate slightly from the desired, input value. It is good practice to generate several candidates and select those closest to the desired scaling. The selected series were then linearly rescaled to have the desired mean and variance. These series were then cumulatively summed to yield a series of onset times, combined with necessary information about offset times and velocities, and written to a MIDI file to be used in Ableton.

matching, there would be no way to ensure observations in the different ITI and IOI series corresponded in terms of the real-time tapping behaviors, no way to accurately calculate the ASY values, and no way to accurately assess the short-term cross-correlations between the series. So, the raw data were first submitted to custom (Matlab) matching algorithms. A detailed description of these algorithms is provided in Appendix A, including a manipulation check analysis regarding the possible effects of the matching procedure on estimates of fractal scaling. In brief, the matching algorithm for synchronization found the nearest observation in some “response” signal (e.g., leader taps) for every observation in some “timing” signal (e.g., metronome beats), and eliminated extraneous observations. The algorithm for syncopation ensured alternation between the timing and response signals, and eliminated extraneous observations. It is also important to note that the matching algorithms were performed both on the leader and metronome (LM) sub-system and the follower and leader (FL) sub-system independently. This choice reflected the uni-directional couplings across the sub-systems and allowed assessment of both the leader’s coordination with the metronome and, separately, the follower’s coordination with the leader.

Following the matching procedure, all of the series were still defined by the onset times. The ITI series were calculated by differencing the series of onset times. The ASY series were calculated by subtracting the onset time of the comparison signal from that of the referent signal for every pair of matched observations. In the case of synchronization, negative ASY values indicated “anticipation”, wherein the event in the comparison signal preceded the referent signal (e.g., the leader tapped slightly before the onset of the metronome). In the case of syncopation, the matching algorithm always began with the first observation in the referent signal, so the sign of the ASY values was always positive.

All these time series still required several standard pre-processing steps before being submitted to the time series analyses (Eke et al., 2000). First, observations within the first ten seconds of recording were removed to eliminate transient behavior. Second, outlier observations were excluded. This was accomplished in two steps: (1) removing observations where the ITI was longer than a standard value (2 s) and (2) removing observations where the ITI was three standard deviations or more from the mean. When an observation met exclusion criteria, the corresponding observations were removed in the paired series. After outlier removal, each series was linearly detrended by subtracting the least-squares regression line from the series. Third, series were zero-centered by subtracting the series mean from every observation. Finally, all series were truncated to 512 observations.

The final, pre-processed series were submitted to two time series analyses². First, detrended fluctuation analysis (DFA) assessed the fractal scaling in the ITI and ASY series with the Hurst exponent (H). For ease of interpretation, the Hurst exponents were transformed into α scaling exponents [$\alpha = 2*(H - 0.5)$], where $\alpha \approx 0$ indicates random, “white” noise pattern, $\alpha \approx 1$ indicates persistent, “pink” noise pattern, and $\alpha \approx -1$ indicates an anti-persistent, “blue” noise pattern of variation. Correlations between α exponents of the coupled series provided an assessment of complexity matching. Second, windowed detrended cross-correlation

²The results of these time series analysis were each validated by additional tests. Most importantly, the fractal scaling results from detrended fluctuation analyses were supported by power spectral density analyses.

(WDCC) assessed the short-timescale interaction between paired time series. Specifically, WDCC was applied to the interval series within each sub-system (e.g., metronome IOI and leader ITI), which is common for studies of complexity matching (e.g., Marmelat & Delignières, 2012), and to the asynchrony series and comparison interval series (e.g., leader-metronome ASY and leader ITI). Together, these analyses reveal how the intervals produced by the response system depend on the intervals of the other system and their mutual asynchrony (e.g., Torre et al., 2013). The outcome of WDCC is a set of correlation coefficients, that scale as Pearson coefficients (i.e., -1 to $+1$), across a number of time lags. For example, a coefficient of .75 at lag $+1$ would indicate a strong positive relationship between observations in one time series and observations one sample later in another time series. For the interested reader, the details of the operation of each of these time series analysis are provided in Appendix B.

The outcomes of these time series analyses, as well as basic measures of the time series³ (i.e., number of outliers, mean, coefficient of variation or standard deviation), were submitted to standard inferential statistics (e.g., ANOVA) to assess the effects of the experimental manipulations; typically a 2 (Coordination Mode: Synchronization vs. Syncopation) $\times$ 2 (Metronome Scaling: Pink vs. White) $\times$ 5 (Magnitude: 0.5% vs. 1% vs. 3% vs. 6% vs. 9%) repeated-measures ANOVA. These tests were performed on the leader-metronome and the follower-metronome sub-systems independently. For all ANOVA tests, when Mauchly's Test of Sphericity indicated an inequality in the variances and/or covariances, a Huynh-Feldt correction was applied to the test values. Given the relatively small sample size and the number of required analyses to fully characterize the data set, it is important to interpret the outcomes of these tests with some reservation. Strong hypothesis testing would require a larger sample with fewer, more targeted tests. The purpose of these tests in the current experiments was more exploratory; to provide some estimate of the experimental effects, to inform modelling efforts, and to generate new hypotheses for future research.

2.2. Results and Discussion

In general, the participants had little trouble performing the task. There were very few outlier observations, both between leader and metronome ($M = 5.62$, $SD = 4.2$) and between follower and leader ($M = 10.35$, $SD = 6.15$). Only one pair performed a single trial for which the final series lengths were less than the prescribed standard length (512 points). These series were zero-padded to the standard length for fractal analysis (Eke et al., 2000).

2.2.1. Basic Time Series Measures—The matching and pre-processing procedures did not alter the average inter-onset interval (IOI) of the metronome series ($M = 799.9\text{ms}$, $SD = 0.52$) or the average inter-tap intervals (ITI) of the leader ($M = 798.74\text{ms}$, $SD = 0.79$) and the followers series ($M = 798.31\text{ms}$, $SD = 0.73$), which were very close to the prescribed period in all of the experimental conditions. Analysis also confirmed that the Magnitude manipulation had the intended effect on the variability of the IOI/ITI series. There were only

³Specifically, the basic measures of the mean and variability of the series were calculated after outlier removal, but prior to detrending and centering.

significant main effects of Magnitude for both the metronome IOI, $F(1.01, 5.05) = 646.11$, $p < .0005$, $\eta_p^2 = .99$, and participant ITI series, $F(4, 20) = 74.63$, $p < .0005$, $\eta_p^2 = .94$. The coefficient of variation (*CoV*) of the ITI series was not, however, perfectly matched to that of the IOI series. As shown in Figure 1, the participant series were always more variable than the metronome and there was not a significant difference between the lowest two magnitudes for either the leaders ($p = .46$) or the followers ($p = .99$). The minimal variation the ITI was approximately 5% *CoV*, and the variation in ITI only began to scale with the metronome at a level of 3% *CoV*, being about 2% higher than the metronome on average.

Analysis of the ASY series showed that the leaders effectively synchronized with the metronome, exhibiting a very small positive mean asynchrony⁴ ($M = 19.03$ ms, $SD = 35.95$), and there were no significant differences across the experimental conditions (all p 's $> .36$). There was a significant effect of Coordination Mode, $F(1, 5) = 479.98$, $p < .0005$, $\eta_p^2 = .99$, on the average asynchrony between followers and leaders. This effect simply reflected the difference between synchronization ($M = -51.27$ ms, $SD = 72.04$), where the followers anticipated the leaders' taps, and syncopation ($M = 303.57$ ms, $SD = 53.38$), where followers taps fell roughly halfway between the leaders' taps.

There were, however, significant differences in the variability of the asynchrony as a function of Magnitude, $F(3.68, 18.4) = 19.53$, $p < .0005$, $\eta_p^2 = .80$, for both sub-systems, and an effect of Metronome Scaling, $F(1, 5) = 9.13$, $p = .029$, $\eta_p^2 = .65$, for the LM sub-system. Specifically, the pattern of standard deviation in ASY series across the magnitude conditions closely matched that of the ITI series. There was very little difference between the 0.5% ($M = 52.05$ ms) and 1% ($M = 50.07$ ms) *CoV* magnitudes, and a steady increase across the 3% ($M = 58.05$ ms), 6% ($M = 67.42$ ms), and 9% ($M = 83.43$ ms) magnitudes. This pattern was not specific to the mode of coordination between leader and follower; the same increase in asynchrony variability was present during both synchronization and syncopation. Finally, the variability in asynchrony between the leader and metronome was slightly greater for the random ($M = 74.4$ ms) than the fractal ($M = 69.92$ ms) metronome conditions.

In summary, the manipulation of the magnitude of variation in the metronome stimulus affected the variability of the participants' tapping behavior, without causing large deviations in the prescribed tapping periods or the expected average asynchrony. Most importantly, participants were not able to match the variability in the metronome at the two lowest magnitude conditions, showing instead a minimal *CoV* near 5%. Also, the variability in the asynchrony of the participants' tapping increased along with the variability in their inter-tap intervals, and the variability in the asynchrony between leader and metronome tended to be slightly higher for the random metronomes. These effects in the basic measures is important background information for interpreting the effects of the experimental manipulations on complexity matching.

⁴Although this mean asynchrony is positive, indicating that leaders tended to lag slightly behind metronome beats, the distribution around this mean did include several cases with negative mean asynchrony ($\approx 29\%$). It is likely that this finding still represents stable synchronization between leader and metronome, but with a small phase shift (e.g., Amazeen, Schmidt, & Turvey, 1995).

2.2.2. Fractal Scaling and Complexity Matching—A manipulation-check analysis confirmed that, after matching and pre-processing, the intended effect of Metronome Scaling was still present, $F(1, 5) = 109.89$, $p < .0005$, $\eta_p^2 = .96$, with fractal metronomes showing the intended, strongly persistent scaling ($M = 0.89$, $SD = 0.11$) and random metronomes showing near random scaling ($M = -0.05$, $SD = 0.16$) across all the experimental conditions. The participants' ITI series, however, showed a very different pattern of fractal scaling across the experimental conditions. There were significant main effects of Magnitude (mean $F(4,20) = 120.26$, mean $p < .0005$, mean $\eta_p^2 = .94$) and Metronome Scaling (mean $F(1,5) = 42.59$, mean $p = .002$, mean $\eta_p^2 = .90$), as well as Magnitude and Metronome Scaling interaction effects (mean $F(2.72, 13.6) = 20.92$, mean $p < .0005$, mean $\eta_p^2 = .81$) for both leader and followers. As displayed in Figure 2, the fractal scaling of the participants' ITI showed different patterns of approach to the scaling of the metronome IOI series for the fractal and the random metronomes.

In the lowest magnitude condition (0.5% *CoV*), the participant ITI series showed the moderate anti-persistent scaling characteristic of coordination with an isochronous metronome (e.g., Chen et al., 1997) for both the random ($M = -0.49$, $SD = 0.22$) and fractal metronomes ($M = -0.39$, $SD = 0.12$). The increase in the scaling of the participant ITI series with increases in magnitude, however, differed with the metronome scaling. For the random metronomes, there was also moderate anti-persistent scaling in the 1% *CoV* magnitude ($M = -0.54$, $SD = 0.11$) and then an abrupt shift to near random scaling for the highest three magnitudes ($M = -0.17$, $SD = 0.12$). In contrast, there was a more gradual shift for the fractal metronomes, with a small increase to near-random scaling at 1% *CoV*, ($M = -0.16$, $SD = 0.19$), another small increase to moderate persistent scaling at 3% *CoV*, ($M = 0.42$, $SD = 0.27$), and a small increase to relatively strong persistent scaling for the highest two magnitudes, ($M = 0.65$, $SD = 0.16$). There was also a significant main effect of Coordination Mode, $F(1, 5) = 6.73$, $p = .049$, $\eta_p^2 = .57$, on the follower's ITI scaling, but no significant interaction between Coordination Mode and Magnitude ($p = .12$), indicating a general tendency for the follower to produce slightly higher scaling values during syncopation than synchronization. It is worth noting that even at the higher magnitudes the maximum scaling of the leaders' ITI ($\alpha \approx 0.75$) never reached the level of the metronome IOI scaling ($\alpha \approx 0.9$). Instead, the maximum leader scaling was commensurate with past studies of self-paced and continuation finger tapping (e.g., Chen et al., 1997).

Interestingly, Marmelat and colleagues (2014) reported a strikingly similar finding regarding the transition to complexity matched behavior. They did not vary the scaling of their pacing stimuli, but rather compared metronomes with persistent scaling ($H = 0.9$; $\alpha = 0.8$) to isochronous metronomes. They noted that one subgroup of participants showed abrupt shifts in their ITI dynamics from anti-persistent scaling to the scaling of the fractal metronome as the magnitude of variation in metronome increased. Another subgroup showed a more gradual increase in their ITI scaling. They explained this by appealing to differences in the ability to perceive inter-beat variation (Ehrle & Samson, 2005; Goldberger et al., 2002). Individual differences in temporal perception are certainly important, but the results of the current experiment suggest that these shifts might also be influenced by the kind of scaling in the pacing stimulus. This effect is most prominent in the difference between the fractal

and random metronome conditions at the 1% *CoV* level of magnitude. One possible explanation of this difference is that even at this level of Magnitude the longer-timescale fluctuations in the fractal metronome presents the participants with coherent, perceptible trends to which they can coordinate their behavior.

To evaluate this possibility, we conducted some secondary analyses in which we carefully examined the scaling plots. As displayed in the average DFA plots for the fractal metronomes (Figure 3; left), a markedly different pattern of scaling characterized shorter- and longer-timescale variation in the participants' ITI dynamics at the 1% *CoV* magnitude level. The shorter-timescale windows (16 to 96 samples) revealed a slightly anti-persistent pattern, whereas the longer-timescale windows displayed a strongly persistent pattern of variation closer to the scaling of the pacing stimulus. In contrast, the plots for the random metronome condition show consistent scaling across all timescales. These results support the idea that the participants may have coordinated to the longer-timescale patterns of variation in the fractal metronome at the 1% *CoV* level of magnitude, taking on its persistent scaling at the longer timescale of their own ITI dynamics. In turn, this effect created different shifts to the scaling of the pacing stimulus between the fractal and metronome conditions. This account is potentially consistent with Marmelat and colleagues (2014) assertion that these different shifts were explained by participants' abilities to perceive the inter-beat variation in the metronome. That is, it is the coherent trends in the fractal metronome IOI (i.e., gradual increases and decreases in the speed of the metronome) that the participants might have been able to perceive, even when the overall magnitude of variation in the stimulus was quite small. It is important state, however, that these results could also be explained by short-term error correction. If the participants maintained coordination with the metronome by making local adjustments, these adjustments could have had cumulative effect over longer periods of measurement, picking up the longer timescale trends in the metronome even though the participants were only sensitive to shorter timescale deviations. Directly testing between these two hypotheses was not possible using the current design, but it is an important goal for future complexity matching research.

It was also important to directly evaluate the correlation between the scaling exponents of the interval series for the pairs of interacting systems, which is how past studies have indexed the strength of complexity matching (e.g., Marmelat & Delignières, 2012; Stephen et al., 2008). The complexity matching plots depicting these relationships, for both the LM and FL sub-systems, are presented in Figure 4. In the current design, however, observations were not independent across the experimental conditions, having come from the same participant pairs. So, to provide a quantitative assessment that addressed these dependencies in the data, separate correlation coefficients were calculated within each condition using the six independent observations (six scaling exponents from one system (e.g., metronome) and six from the paired system (e.g., leader) used in a correlation, within each of the twenty experimental conditions)⁵. Based on the previous results, a basic question was whether the correlations would differ between the lower two magnitude conditions (0.5% and 1% *CoV*),

⁵We also performed these correlation tests using scaling values derived from the shorter- and longer-timescale regions separately, in line with the procedure from Marmelat et al., (2014). We found very similar patterns of complexity matching correlations within the shorter- and longer-timescale values. The fit of the regression line to the fluctuation functions was generally very strong, both when

where the participants' tapping intervals had not matched to the scaling of the metronome, and the higher magnitudes (3%, 6%, and 9% *CoV*). Within the LM sub-system, the correlations between the scaling of the metronome IOI and leader ITI series were generally weaker and more variable in the lower two magnitudes (mean $r = .28$, $SD = .39$, range = $-.36$ to $.84$) than in the higher three magnitudes (mean $r = .84$, $SD = .08$, range = $.65$ to $.95$). This discrepancy was much smaller within the FL sub-system, with strong positive correlations for both the lowest two magnitudes (mean $r = .75$, $SD = .20$, range = $.45$ to $.997$) and the higher three magnitudes (mean $r = .86$, $SD = .23$, range = $.16$ to $.98$).

These tests suggested one important caveat to the conclusions from the categorical analyses of the condition mean scaling values. The results of categorical analysis implied that complexity matching does not emerge below a minimal magnitude of variation. These continuous analyses support this conclusion in terms of the leaders' matching to the metronome stimulus; with a relatively weak and unstable relationship between leaders' ITI and metronome IOI scaling in the lowest two magnitudes. The followers' ITI scaling was strongly correlated to that of the leaders across all of the magnitude conditions. This is likely due to the fact that the amount of variation in the leaders' ITI series, even in the lowest magnitude condition ($\approx 5\%$ *CoV*), substantially exceeded the estimated minimal 3% *CoV* level needed for complexity matching to emerge. It is also important to consider that the follower-leader relationship is defined by continuous visual coupling, while the leader-metronome relationships is defined by discrete auditory coupling. Further research directly comparing auditory and visual coupling to fractal timing signals would be required to resolve this issue.

In addition to these tests of complexity matching between the inter-event intervals of the interacting systems, we also evaluated the fractal scaling of the asynchrony series. There was a significant effect of Magnitude, $F(4, 20) = 13.47$, $p < .0005$, $\eta_p^2 = .73$, on the ASY scaling within the LM sub-system. The scaling of the asynchrony between leader and metronome steadily decreased from moderate ($M = 0.7$, $SD = 0.25$) to weak ($M = 0.3$, $SD = 0.29$) persistent scaling from the lowest to the highest magnitude conditions (see Figure 5; left). There were no significant effects involving Magnitude within the FL sub-system (all p 's $> .27$), but there was a significant effect of Coordination Mode, $F(1, 5) = 32.38$, $p = .002$, $\eta_p^2 = .87$, with synchronization producing weak ($M = 0.24$, $SD = 0.25$) and syncopation producing moderate persistent scaling ($M = 0.68$, $SD = 0.37$) across all the other experimental conditions (Figure 5; right). It is interesting to note that there were no effects involving Metronome Scaling on the ASY scaling ($p = .40$), indicating that the scaling of the asynchrony was not influenced by the scaling present in the metronome stimulus. It is important to contextualize these findings within the findings of the previous analyses. Recall, within the LM sub-system, where the leader was always synchronizing to the metronome, increases in the magnitude of the metronome variation were associated with an increase in the variability of the leader-metronome asynchrony. These analyses additionally show that as the asynchrony became more variable, its fractal scaling also shifted toward

including the full range of bin sizes ($R^2 = .95$) and when the largest five bin sizes were dropped ($R^2 = .96$), and the average absolute deviation in the estimated Hursts was small ($M = 0.05$, $SD = 0.035$). So, only the Hursts estimated from the all scales are reported here.

more random variation. Despite the increasing variability and randomness of the asynchrony series in the higher three metronome conditions, there was very strong complexity matching between the leaders' ITI series and the metronome IOI series. Similarly, within the FL sub-system, there was relatively strong complexity matching between the followers and leaders during both synchronization (mean $r = .86$, $SD = .23$, range = .16 to .95) and syncopation (mean $r = .77$, $SD = .20$, range = .45 to .997). In contrast, there was a clearly difference in the fractal scaling of their asynchrony series between the coordination modes. These differences between the scaling evident in the two interval series and the scaling of the asynchrony series must be considered along with the results of the other analyses to understand the coordination underlying complexity matching.

In summary, these analyses confirmed that the experimental manipulations affected the fractal scaling of the participants tapping intervals and the complexity matching between the interval series, as well as the scaling of the asynchrony series. First, consistent with the past research (Marmelat et al., 2014), these results support the hypothesis there is a minimal level of magnitude in the variation of the pacing stimulus required for complexity matching to emerge. However, the current results also showed that the scaling of the leaders' ITI was, in fact, influenced by the scaling of the metronome below this minimal level. At the lowest level of magnitude (0.5% *CoV*), the leaders' ITI showed the anti-persistent scaling characteristic of coordination with isochronous metronomes (Chen et al., 1997; Torre & Delignières, 2008), regardless of the actual scaling in the metronome IOI series (fractal vs. random). However, at the 1% *CoV* level, the leaders' ITI series displayed *partial complexity matching*, in which the long-timescale variation in the leaders' ITI was closer to the metronomes' scaling. Second, unlike the leader-metronome matching, the follower-leader matching was consistently strong across all levels of magnitude. This effect is likely due to the fact that the smallest amount of variation in the leaders' tapping behavior exceeded the minimum level needed to elicit the emergence of complexity matching ($\approx 5\%$). Finally, with respect to the primary research question regarding the coordination involved in complexity matching, the results did not suggest that coordination mode strongly influenced complexity matching between the participants. Recall, syncopation was expected to produce weaker coordination than was synchronization. Although complexity matching was slightly stronger during synchronization, there was also strong complexity matching during syncopation, and the coordination mode manipulation did not tend to interact with the other experimental factors. In contrast, the coordination mode did strongly influence the fractal scaling of the asynchrony series, with syncopation producing much more persistent scaling than synchronization. This dissociation between the effects of coordination mode on the scaling of the interval series (i.e., complexity matching) and the asynchrony series was potentially important for better understanding the complexity matching effect, because these task variables were interrelated. It was important, however, to further examine these differences in the scaling were associated with differences in the patterns of short-term cross-correlation between these task variables.

2.2.3. Short-term Cross-correlation—The WDCC analyses revealed the pattern of short-term cross-correlation between the interval series generally expected from finger tapping tasks (e.g., Coey et al., 2015). For both sub-systems, the peak correlation tended to

fall at lag +1, with much weaker correlations at other time lags. A peak at lag +1 meant observations in the response series (e.g., leader's ITI) were most strongly related to the *preceding* observations in the timing series (e.g., metronome IOI). One exception was that the peak was located at lag 0 for the FL sub-system during syncopation. This indicated the follower's current ITI was most strongly related to the leader's *current* ITI, however, that statement requires qualification. Due to the matching algorithm for syncopation, the followers' "current" ITI was defined by a tap occurring *after* (≈ 400 ms) the leaders' tap *in real time*. To account for this difference in further analyses of the WDCC functions, the time lags were redefined in reference to the location of the peak. For instance, "peak +1" referred to an actual time lag (in sample numbers) of +2 for the synchronization condition, but lag +1 for the syncopation condition.

The WDCC functions for the LM sub-system suggested that the magnitude and fractal scaling of the metronome affected the short-term coordination between leader and metronome (see Figure 6, top row). Analysis confirmed that there were significant effects of Magnitude, $F(4, 20) = 139.95$, $p < .0005$, $\eta_p^2 = .97$, and Metronome Scaling, $F(1, 5) = 51.73$, $p = .001$, $\eta_p^2 = .91$, as well as a trend for an interaction of Magnitude and Metronome Scaling, $F(4, 20) = 2.46$, $p = .079$, $\eta_p^2 = .33$. The lowest two magnitudes produced negligible peak correlations, while the higher three conditions produced increasingly stronger, positive peaks. The random metronomes produced overall stronger peak correlations than did the fractal metronomes, and, in particular, the peak was stronger for the random metronome at the 3% magnitude condition ($p = .013$). Analysis also confirmed significant effects of Magnitude, $F(4, 20) = 7.46$, $p = .001$, $\eta_p^2 = .60$, and Metronome Scaling, $F(1, 5) = 35.67$, $p = .002$, $\eta_p^2 = .88$, and a significant interaction of Magnitude and Metronome Scaling, $F(4, 20) = 7.05$, $p = .001$, $\eta_p^2 = .59$, at the peak +1 time lag. Here, the fractal metronomes produced small cross-correlations between leader and metronome at the higher three magnitudes ($r \approx 0.2$) than did the random metronome condition ($r \approx 0.05$).

The WDCC functions for the FL sub-system also suggested that the magnitude and metronome scaling manipulations affected the cross-correlation between the followers and leaders (see Figure 6; bottom row). Analyses of the peak correlations confirmed a significant effect of Magnitude, $F(4, 20) = 12.19$, $p < .0005$, $\eta_p^2 = .71$, but there were no effects involving Metronome Scaling or Coordination Mode (all p 's $> .29$). The strength of the peak correlation increased with increases in magnitude, but, unlike the LM sub-system, the peaks at the lowest magnitudes were considerable ($r \approx 0.5$). Analyses also showed significant effects of Magnitude, $F(4, 20) = 7.91$, $p = .001$, $\eta_p^2 = .61$, and Metronome Scaling, $F(1, 5) = 40.39$, $p = .001$, $\eta_p^2 = .89$, as well as an interaction of Magnitude and Coordination Mode, $F(4, 20) = 3.13$, $p = .037$, $\eta_p^2 = .39$, at the peak +1 time lag. As in the LM sub-system, the fractal metronome produced stronger peak +1 correlations between the follower and leader than did the random metronome condition. There were also stronger peak +1 correlations for greater magnitude conditions, with the synchronization mode producing slightly weaker correlations ($r \approx .01$) than the syncopation mode ($r \approx .08$) in the highest, 9% *CoV* magnitude condition.

The WDCC analyses of the short-term correlations between the asynchrony and interval series revealed a consistent pattern in both sub-systems, with weak, positive peak

correlations at lag 0 and strong, negative correlations at lag +1 (see Figure 7). This pattern is consistent with a 'corrective' interaction between the interacting systems, in which the current interval is corrected based on the preceding asynchrony (see Delignières et al., 2008). That is, the negative correlation at lag +1 indicated that shorter intervals were associated with larger previous asynchrony values, so when the participant lagged behind the metronome (or their partner) on the previous tap, they tended to shorten their interval on the next tap. The small positive correlation at lag 0 indicated that longer intervals yielded a slight tendency to lag on the current tap, producing a larger (positive) asynchrony value.

The WDCC functions for the LM sub-system closely resembled those of the FL subsystem during the synchronization condition (Figure 7; top row), and suggested that both the magnitude and fractal scaling of the metronome affect the short-term coordination between the leaders' intervals and the leader-metronome asynchrony. Analysis of the lag +1 correlations confirmed significant effects of Magnitude, $F(1.67, 8.35) = 57.29$, $p < .0005$, $\eta_p^2 = .92$, and Fractal Scaling, $F(1, 5) = 62.41$, $p = .001$, $\eta_p^2 = .93$, and a significant interaction effect between Magnitude and Fractal Scaling, $F(3.89, 19.48) = 3.89$, $p = .018$, $\eta_p^2 = .44$. The greater magnitude conditions lead to stronger negative peaks, the random metronome lead to slightly stronger negative peaks than did the fractal metronome, particularly for the lower magnitude conditions. Also, analysis of the lag 0 correlations confirmed an effect of Magnitude, $F(3.39, 16.96) = 128.7$, $p < .0005$, $\eta_p^2 = .96$, where greater magnitude produced weaker positive peak correlations.

The WDCC functions of the FL sub-system suggested a large difference between the synchronization and syncopation conditions (see Figure 7; bottom row). Analysis of the lag +1 correlations confirmed an effect of Coordination Mode, $F(1, 5) = 235.84.18$, $p < .0005$, $\eta_p^2 = .98$, an interaction of Coordination Mode and Fractal Scaling, $F(1, 5) = 17.88$, $p = .008$, $\eta_p^2 = .78$, and an interaction of Coordination Mode and Magnitude, $F(3.23, 16.2) = 12.54$, $p < .0005$, $\eta_p^2 = .72$. There were also effects of Magnitude, $F(2.11, 10.52) = 18.19$, $p < .0005$, $\eta_p^2 = .78$, and Coordination Mode, $F(1, 5) = 13.12$, $p = .015$, $\eta_p^2 = .72$, for the lag 0 correlations. Overall, the synchronization condition produced the same results as observed in LM sub-system. The greater magnitude conditions produced weaker positive lag 0 and stronger negative lag +1 correlations, and the random metronome condition produced stronger negative correlations than did the fractal metronome condition. The syncopation condition also showed similar effects of the magnitude and scaling of the metronome, but the peak correlations were overall much weaker than those in the synchronization condition. This large difference between the two coordination modes indicated that the followers' inter-tap intervals were much less dependent on the follower-leader asynchrony during syncopation.

In summary, the analyses of the short-term cross-correlation showed that the intervals produced by the response systems were related to both the intervals of the timing systems and their mutual asynchronies. More importantly, these analyses revealed that the strength of the peak correlations was affected by the experimental manipulations in certain ways that reflected the observed effects on fractal scaling and complexity matching. Recall, the fractal analysis of the interval series showed that the leaders' ITI scaling was weakly matched to the scaling of the metronome at the lowest two magnitudes and much more strongly matched at

the higher three magnitudes, while the scaling of the followers' ITI was strongly matched to the leaders' at all levels of magnitude. Moreover, the leaders' scaling tended to more gradually come to match that of the metronome with increases in magnitude for the fractal metronomes, but abruptly shifted to the scaling of the metronome for the random metronomes. Very similar effects were observed in the peak correlations between the paired interval series. The followers' current ITI was strongly correlated to the leaders' previous ITI at all levels of magnitude, while the leaders' current ITI was increasingly correlated to the previous metronome IOI with increases in magnitude, more gradually for fractal metronomes than for random metronomes. Similarly, the effect of coordination mode on the fractal scaling of the asynchrony series was mirrored in the short-term cross-correlations between the follower's intervals and the follower-leader asynchrony, with the syncopation mode producing weaker peak cross-correlations and more persistent fractal scaling. Together, these findings suggested that differences in the fractal scaling of the task variables and complexity matching were related to the strength of the "local", short-term interactions.

2.3 Conclusion

There were a number of interesting findings in Experiment 1 that might speak to the primary research question regarding the coordination involved complexity matching. First, the manipulation of coordination mode was expected to produce differences in the strength of coordination between participants. This was not evident in the cross-correlations of the two interval series. The peak correlations were roughly equal in syncopation and synchronization, suggesting equally strong coordination in both modes. Complexity matching between the interval series was also roughly equal in syncopation and synchronization. Coordination mode did, however, seem to strongly influence the follower-leader asynchrony dynamics. Consistent with the past research (Chen et al., 2001) the syncopation mode produced much more persistent asynchrony scaling than did synchronization. A similar effect was evident in the cross-correlations, with the followers' tapping intervals being much less related to their asynchrony to the leaders' taps during syncopation than synchronization. So, the asynchrony series provide a different picture of the coordination involved in the task performance than do the interval series. This is particularly interesting given that it might be the asynchrony dynamics which better quantify the performance (Chen et al., 1997). Together, these findings suggest that complexity matching can be equally strong in conditions that produce very different coordination dynamics.

Second, it is important to note that the results suggested a strong association between patterns in the short-term cross-correlations and patterns in the fractal scaling and complexity matching. This association can be taken as support for the conclusion that complexity matching is driven by "local" coordinative process, but this does not necessarily exclude the possibility of some other "global" coordinative process. The global coordination account does not necessarily assert that changes in the local, beat-to-beat coordination (i.e., interval cross-correlation) should not be related to changes in complexity matching. Two systems might be *coupled* across multiple scales and still realize a behavior that exhibits strong "local" coordination, and it is possible that changes in both the cross-correlations and

complexity matching might be caused by changes in the multiscale interactions between systems (Stephen & Dixon, 2011).

Third, the above considerations make it clear that a critical difference between the “local” and “global” accounts of complexity matching is the nature of the *coupling* between the systems. The “partial complexity matching”, wherein leaders synchronizing with fractal metronomes started to produce persistent scaling in their longer timescales at a low magnitude (1% *CoV*), might speak to this question. It is possible that this result reflects a perceptual coupling at these longer timescales (Marmelat et al., 2014), with participants being able to detect the larger fluctuations in the metronome that occur at these timescales. Alternatively, it is possible that these findings reflected an accumulative result of local coordination with the metronome. That is, weak local coordination (evidenced by the cross-correlation) could have caused the participants’ tapping behavior to match the longer timescale patterns in the metronome without matching the shorter timescale fluctuations well. The design of the current study was not able to test between these two hypotheses. These results did support that the exact degree of persistence in the metronome might be another important experimental factor, which we explored further in Experiment 2.

3. Experiment 2

One of the main findings of Experiment 1 suggested that the participants might be perceptually attuned to the longer-range oscillatory patterns embedded in the fractal metronomes. If the participants were perceptually coupled via these long-range patterns, a further hypothesis was that increasing the persistent scaling in the metronome would lead to stronger levels of complexity matching. In their second experiment, Marmelat and colleagues (2014) explored different degrees of scaling in the metronome, including moderately anti-persistent ($\alpha = -0.6$), random ($\alpha = 0$), mildly persistent ($\alpha = 0.2$), and strong persistently ($\alpha = 0.8$) scaling, as well as isochronous and self-paced conditions. They found the strength of complexity matching was greatest when the persistent scaling in the metronome was highest (i.e., $\alpha = 0.8$). This could have resulted because the metronome was producing a signal with scaling very near the participants’ own, “intrinsic” degree of scaling (Kello & Van Orden, 2009; Van Orden et al., 2011). It is also possible that this result could have reflected a more general effect of increasing the persistence in the metronome. That is, greater persistence in the metronome onsets might have provided more perceptually coherent trends to which the participants could coordinate, and this effect might extended past the range of fractal scaling typically defining uncoupled, self-paced tapping behavior ($\alpha \approx 0.75$). However, the Experiment 1 findings that participants ITI scaling never reached the scaling in the metronome IOI ($\alpha \approx 0.75$) suggests there might be some constraints on the range of scaling in the timing stimulus that can yield stable complexity matching. The purpose of Experiment 2 was to test these ideas, by examining metronomes with even greater degrees of persistent scaling than those employed by Marmelat and colleagues (2014). We chose to include metronomes with mildly anti-persistent ($\alpha = -0.35$), random ($\alpha = 0$), mildly persistent ($\alpha = 0.35$), and strongly persistent scaling ($\alpha = 0.7$), but also included two more conditions with even greater degrees of persistence ($\alpha = 1.05$ and $\alpha = 1.4$), to try to determine which condition would yield the strongest complexity matching. As in

Experiment 1, we also considered how these manipulations would affect the short-term correlations between the paired time series and the scaling of the asynchrony series.

3.1. Method

3.1.1. Participants—A total of 14 graduate student volunteers from the Department of Psychology at the University of Cincinnati participated in the experiment. Two of these participants had participated in Experiment 1, but were paired with a different partner and were assigned to the role opposite to the one they completed in Experiment 1. As the study required multiple sessions, all participants were assigned to pairs ($N = 7$) according to their respective availabilities. The mean age of the participants was 28.0 years, 71.4% of the participants were male, and 100% of the participants were right-handed.

3.1.2. Apparatus—As before, the finger tapping was recorded using USB MIDI keyboards (± 5 ms error) connected to a PC computer running Ableton Live (v. 8.2) software (Ableton, Berlin Germany), which both recorded the participants' taps and presented auditory metronome stimuli to the participants through pairs of headphones.

3.1.3. Design—As in Experiment 1, the metronome stimuli all had an average inter-onset interval (IOI) of 800 ms, but here the metronomes all had the same amount of variation around the mean IOI ($CoV = 4.5\%$; $SD = 36$ ms). The level of 4.5% CoV was selected based on the results of Experiment 1. Recall, at the level of 3% CoV the leaders had begun to complexity match to the metronome, and so we chose a value higher than this minimum in order to ensure relatively stable complexity matching. However, the level of 6% CoV was not selected because very stable complexity matching might have obscured the possibly more subtle effects of varying the scaling of the metronome. The metronome scaling was manipulated within-subjects, and each pair experienced six metronomes, ranging from mildly anti-persistent ($\alpha \approx -0.35$) to very strongly persistent ($\alpha \approx 1.4$) IOI series in small steps ($\alpha = 0.35$). However, each pair of participants experienced their own unique set of metronome stimuli, and the scaling of each set was allowed to slightly deviate from the categorical values (i.e., $\alpha = -0.35, 0.0, 0.35, 0.7, 1.05$, and 1.4). More specifically, the exact scaling of the metronome IOI series was allowed to deviate from the categorical values by approximately ± 0.15 . The rationale for this decision was to obtain a more fine-grained picture of the relationship between the metronome scaling and the strength of complexity matching. The limitation of this decision was that it confounded the pair-level individual differences with differences in the metronome scaling for the categorical analyses⁶. This is a potential design limitation, but, given the aims of the study, it seemed important to provide a more densely populated range of metronome scaling values to be used in a continuous assessment of complexity matching, like those presented in the earlier complexity matching plots (Figure 4). Although the previous findings suggested that the $\alpha = 0.7$ condition might produce strongest matching, it was possible that this would occur at some other, non-

⁶Analyses using outcomes adjusted for the differences of the actual metronome scaling from the categorical value supported the conclusions from the unadjusted analyses. Specifically, for each trial, the difference between the actual and categorical metronome scaling values was calculated. The outcome for that trial was then regressed on these difference scores, and the “residualized” outcomes were finally used in ANOVA tests with Metronome Scaling as a categorical variable. The pattern of results for the adjusted data was the same for all outcomes, with one minor exception, which is indicated below.

represented level of metronome scaling (e.g., $\alpha = 0.5$). Each pair also performed a self-paced leader condition to provide an estimate of the strength of complexity matching when the leader was unconstrained by a timing stimulus. As before, the coordination between leader and follower was manipulated within-subjects and every pair performed both blocks of synchronized and syncopated tapping trials. Half of the pairs performed their synchronization trials first, and the other half performed syncopation first. This design resulted in a grand total of 14 trials for every pair. Again, the metronomes consisted of series of 600 beats, each trial lasting about 8 minutes.

3.1.4. Procedure—Following informed consent, the participants were randomly assigned to the role of leader and follower (except for the two participants mentioned above) and were seated across a table from one another, facing their keyboards. The leaders were instructed that they would always attempt to tap in synchrony with the metronome, except during the self-paced trial. During the self-paced trial, they were told that they would hear 10 seconds of a metronome beat, that they should synchronize their taps to the metronome, and that they should continue to tap after the metronome discontinued, attempting to maintain the same frequency of tapping. Depending on the order of the coordination modes for that pair, the follower was either told that they would tap in synchrony or syncopation with the leader. Within each block of seven trials, the pair would always complete the unconstrained, self-paced tapping trial first, and then the remaining six trials with metronomes of different scaling values in a randomized order. When the pair had completed their first block of trials, the coordination mode between the participant switched, and the follower was instructed to now tap in the other pattern with the leader. As in Experiment 1, participants were allowed short breaks between trials and completed the experiment across 3 to 4 depending on their availability.

3.1.5. Data Analysis—The exact same data analysis procedures employed in Experiment 1 were employed in this second experiment.

3.2. Results and Discussion

As in Experiment 1, the participants generally had little trouble with the task. One pair was excluded from further analysis because the leader consistently had trouble maintaining synchrony with the metronome. Of the remaining six pairs, there were only six trials where the length of the final, pre-processed series was shorter than the standard 512 observations. Five trials had lengths near enough to the standard length ($M = 497.6$, $SD = 11.72$, range = 476 – 506) so that they could be zero-padded to 512 observations without compromising the validity of the fractal analyses. The sixth, however, was very short of the standard (268 observations). In this case the leader participant frequently did not release the key entirely, resulting in very many missed observations and extremely long ITI values. Rather than exclude the pair entirely, their score for this condition (i.e., self-paced synchronization) was replaced by mean imputation.

3.2.1. Basic Time Series Measures—As in Experiment 1, the average metronome IOI and participant ITI were very close to the prescribed period (800 ms) after matching and pre-processing. Also as before, the participants' ITI series were generally more variable (SD

≈ 60 ms) than the metronome IOI series ($SD \approx 37$ ms). For the FL sub-system, there were no significant differences in the variability of the participant ITI (all p 's > .22). There was, however, an effect of Metronome Scaling, $F(5, 25) = 6.13$, $p = .001$, $\eta_p^2 = .55$, within the LM sub-system, where the leaders' ITI slightly increased for the metronomes with more persistent scaling. In terms of absolute magnitude, however, this effect was not very large (range = 34.69 – 75.39 ms), and it is unlikely this issue had significant ramifications for interpreting the outcomes of the other analyses.

Analysis of the ASY series showed that the leader successfully synchronized with the metronome, with a negative mean asynchrony ($M = -36.06$ ms, $SD = 54.27$), indicating that the leaders' taps tended to anticipate the beats of the metronome. There were no significant differences in the average leader-metronome asynchrony across the experimental manipulations (all p 's > .28). There was a significant effect of Coordination Mode on the average asynchrony within the follower-leader system, $F(1, 5) = 775.11$, $p < .0005$, $\eta_p^2 = .99$, where the follower anticipated the taps of the leader ($M = -51.29$ ms, $SD = 44.28$) during synchronization and the followers' taps falling near halfway between the leaders' during syncopation ($M = 350.09$ ms, $SD = 37.91$). There was also a significant effect of Metronome Scaling, $F(6, 30) = 2.68$, $p = .033$, $\eta_p^2 = .35$, where the average asynchrony in the self-paced condition tended to be slightly higher than in the metronome conditions, being nearer to 0 ms during synchronization ($M = -22.98$ ms, $SD = 52.09$) and nearer to half the prescribed period during syncopation ($M = 385.95$ ms, $SD = 57.31$). Lastly, analysis showed that there were no differences in the standard deviation of the asynchrony for either the leader-metronome ($M = 75.76$ ms, $SD = 18.98$) or the follower-leader ($M = 69.88$ ms, $SD = 26.63$) sub-systems (all p 's > .24).

In summary, the analyses of the basic time series measures confirmed that the participants successfully performed the task and that there were no concerning deviations in the average or variability of the intervals and asynchronies across the experimental conditions.

3.2.2. Fractal Scaling and Complexity Matching—First, it was important to examine the scaling of the “final” metronome IOI series, as these final series reflect both the result of the matching and pre-processing procedures and the intended variation in the scaling across the six different sets of metronome stimuli. A regression revealed the scaling of these final IOI series tended to slightly underestimate the values ascribed to the categorical conditions for the metronome scaling (i.e., -0.35, 0, 0.35, 0.7, 1.05, and 1.4). Nonetheless, the range of final IOI scaling came very close to covering the intended range, providing a densely populated range of actual metronome scaling from moderate anti-persistence to very strong persistence (-0.51 to 1.33).

Analysis of the leaders' ITI scaling from the LM sub-system showed very strong complexity matching to the metronome IOI scaling (see Figure 8). As in Experiment 1, the leaders' ITI scaling was systematically lower than that of the metronome IOI series, only reaching a maximum value of $\alpha \approx 1$ in the highest metronome scaling condition, but nonetheless strongly correlated to the scaling of the IOI series ($r \approx .98$). The correlation between the followers' and leaders' ITI scaling was slightly weaker ($r \approx .90$) than between the leader and metronome, but still revealed very strong complexity matching. More importantly, there was

a very similar level of complexity matching in both the synchronization ($r = .91$) and syncopation ($r = .92$) conditions. However, the exact linear relationship between the leaders' and followers' ITI scaling did differ between the two coordination modes, with the syncopation condition yielding a smaller slope (0.78) than the synchronization condition (0.86). A follow-up analysis of the average ITI scaling revealed a significant interaction between Participant Role (leader vs. follower) and Metronome Scaling, $F(6, 30) = 3.03$, $p = .02$, $\eta_p^2 = .38$, where the follower tended to produce slightly lower scaling values than the leader, especially in the more persistent metronome conditions of $\alpha = 0.7$ ($p = .02$), $\alpha = 1.05$ ($p = .02$), and $\alpha = 1.4$ ($p = .001$). This analysis also revealed a significant interaction of Coordination Mode and Metronome Scaling, $F(6, 30) = 3.33$, $p = .01$, $\eta_p^2 = .40$, where both the leader and follower produced lower scaling values for their self-paced ITI in syncopation ($M = 0.39$, $SD = 0.21$) than in synchronization ($M = 0.69$, $SD = 0.11$) and slightly higher scaling values for the $\alpha = 0$ metronome scaling condition during syncopation ($M = 0.10$, $SD = 0.10$) than during synchronization ($M = 0.05$, $SD = 0.11$). This latter effect accounts for the lower overall slope in the syncopation condition.

To evaluate the primary question of what level of metronome scaling produced the strongest complexity matching, we conducted Breusch-Pagan tests of heteroscedasticity, which assessed the homogeneity of residual variance around the regression line within the complexity matching plots. For the LM sub-system, there was no significant heteroscedasticity in predicting the leaders' ITI scaling from the metronome scaling, $\chi^2(1) = 0.013$, $p = .91$. For the FL subsystem, however, the synchronization condition did yield significant heteroscedasticity, $\chi^2(1) = 7.97$, $p = .005$, while the syncopation condition did not, $\chi^2(1) = 1.10$, $p = .294$. Figure 9 displays the absolute residual variance as a function of the referent systems' scaling. Compared to the FL sub-system, there was very little residual variance within the LM sub-system, consistent with the observation of very strong complexity matching between the leader and the metronome. In the FL sub-system there was considerably more residual variance, and that residual variance was not homogeneously distributed, being greater when the leaders' ITI displayed anti-persistent and random variation and lesser when the leaders' ITI displayed more persistent variation. As the leaders' ITI scaling was limited to a maximum value of $\alpha \approx 1$, these tests could not evaluate the strength of the complexity matching at even more persistent levels. However, visual examination of the plot did show that the residual variation was least when the leaders' ITI was near the range of scaling commonly observed in self-paced tapping ($\alpha \approx 0.6 - 0.8$), and increased slightly when the leaders' ITI was more persistent. This scaling range was confirmed by the average participant ITI scaling during self-paced synchronization ($M = 0.69$, $SD = 0.12$). In contrast, the residual variation was relatively homogeneously distributed when the follower and leader tapped in syncopation.

Finally, the analyses of the fractal scaling of the ASY series generally showed the expected pattern of results. For the LM sub-system, the asynchronies showed a moderate persistent scaling ($M = 0.56$, $SD = 0.11$), and there were no significant differences across the experimental conditions (all p 's $> .23$). Within the FL sub-system, there was the expected significant effect of Coordination Mode, $F(1, 5) = 31.25$, $p = .003$, $\eta_p^2 = .86$, with syncopation ($M = 0.58$, $SD = 0.11$) yielding more persistent asynchrony scaling than synchronization ($M = 0.27$, $SD = 0.13$). There was also an interaction of Coordination Mode

and Metronome Scaling, $F(1, 5) = 2.6$, $p = .038$, $\eta_p^2 = .34$. As shown in Figure 10, the lowest five levels of metronome scaling produced the same degree of asynchrony scaling as observed in Experiment 1, with synchronization yielding only mild persistence ($\alpha \approx 0.2$) and syncopation yielding stronger persistence ($\alpha \approx 0.6$). However, the asynchrony scaling within the synchronization condition markedly increased for the $\alpha = 1.4$ metronome scaling condition ($\alpha \approx 0.45$) and for the self-paced condition ($\alpha \approx 0.55$), and these values were not significantly different than those in the syncopation condition (p 's = .25 and .42, respectively).

In summary, the leaders' ITI scaling was very strongly correlated to the IOI scaling across the full range ($\alpha \approx -0.5$ to 1.3), but tended to underestimate the exact value for the most persistent IOI conditions, reaching a maximum scaling of $\alpha \approx 1$. Similarly, the followers' ITI scaling was also strongly correlated to the leaders' ITI scaling, but again tended to underestimate the leaders' scaling, especially in the higher metronome scaling conditions. Finding that response systems systematically yield lower fractal scaling than their timing systems is consistent with other recent findings (Washburn et al., 2015) showing that response systems also tend to be less chaotic than their timing systems.

More interestingly, the tests of heteroscedasticity suggested that certain degrees of metronome scaling produced stronger complexity matching. Between leader and metronome there was slightly weaker complexity matching for the anti-persistent and random metronomes, but this was not great enough to produce significant heteroscedasticity. This finding is in contrast to other studies that have shown considerably weaker complexity matching under these conditions (e.g., Coey, Washburn, & Richardson, 2014; Marmelat et al., 2014), and might reflect an overall very strong complexity matching between leader and metronome in the present experiment. Indeed, between the follower and leader, where the complexity matching was weaker overall, there was significant heteroscedasticity during synchronization tapping. Followers tended to show stronger matching when the leaders' intervals were more persistent, and perhaps showed strongest matching when the leaders were near the scaling value of $\alpha \approx 0.7$ characterizing self-paced tapping. Interestingly, this effect was absent in the syncopation condition, where there was relatively equal degree of matching across all of the leaders' ITI scaling values, even though the overall strength of matching was equal between syncopation and synchronization ($r \approx .90$). This might reflect important differences in the coordination underlying complexity matching between the synchronization and syncopation modes.

Lastly, the analyses of the asynchrony scaling were also in line with the findings of Experiment 1. The scaling in the metronome intervals did not significantly affect the scaling of the leader-metronome asynchronies, although it tended to be slightly higher when the metronome was most persistent. The mode of coordination did affect the follower-leader asynchrony, with syncopation yielding more persistent asynchronies than did synchronization (Chen et al., 2001). However, like the leader-metronome sub-system, the synchronization condition tended to produce slightly higher follower-leader asynchrony scaling when the leaders' intervals were most persistent, as well as in the self-paced tapping condition. Past research has suggested that the coupling between the movement parameters (e.g., stride time and stride length) that characterizes coordination with a persistent, fractal

timing stimulus is considerably weaker in self-paced conditions (Roerdink, Daffertshofer, Marmelat, & Beek, 2015). The current results are consistent with this explanation, and might further suggest that levels of metronome scaling exceeding the range of characteristic scaling of participants' behavior also produce a breakdown of the coupling between key movement parameters. As before, the most relevant question was whether these differences in the fractal scaling and complexity matching were related to differences in the coordination between the interacting systems.

3.2.3 Short-term Cross-correlation—The WDCC functions revealed the same basic patterns as in Experiment 1. The cross-correlation between the paired inter-event intervals revealed a peak correlation at lag +1 for cases of synchronization. As before, the peak correlation was shifted to lag 0 for cases of syncopation. The cross-correlations for the asynchrony-interval pairs revealed a small, positive peak at lag 0 and a stronger, negative peak at lag +1. The subsequent analyses tested for differences in the strength of these peak correlations across the experimental conditions.

The WDCC functions for the paired interval series from the LM sub-system suggested that the higher metronome scaling conditions produced weaker peak correlations. Analysis confirmed a significant effect of Metronome Scaling, $F(5, 25) = 3.74$, $p = .012$, $\eta_p^2 = .43$, with the $\alpha = 1.4$ metronome scaling condition yielding smaller peaks than the other conditions. Unlike the results of Experiment 1, the WDCC functions also suggested the presence of substantial correlations at longer time lags (see Figure 11), with greater metronome scaling yielding larger correlations between leader ITI and metronome IOI. Analysis confirmed significant effects of Metronome Scaling at peak +1, $F(5, 25) = 17.57$, $p < .0005$, $\eta_p^2 = .78$, and peak +2, $F(5, 25) = 5.83$, $p = .001$, $\eta_p^2 = .54$. Together, these findings showed that the very persistent metronome scaling conditions produced a wider distribution of cross-correlation between leader and metronome, with smaller peak correlations, but also larger, longer-range correlations.

The WDCC functions between the interval series within the FL sub-system revealed a similar pattern of effects. Similar to the LM sub-system, there were smaller peak correlations for the highest metronome scaling ($\alpha = 1.4$) and the self-paced conditions (Figure 11). Analysis confirmed a significant effect of Metronome Scaling, $F(6, 30) = 3.67$, $p = .008$, $\eta_p^2 = .42$, with markedly low peaks for self-paced condition. It is also worth noting that self-paced syncopation ($M = 0.21$, $SD = 0.15$) produced an even lower peak correlation than self-paced synchronization ($M = 0.34$, $SD = 0.25$), and this difference was marginally significant ($p = .06$). Unlike the LM sub-system, there were no substantial longer-range correlations.

The WDCC functions between the leaders' ITI and the leader-metronome ASY suggested there were small differences in the strength of the lag 0, positive and the lag +1, negative peak correlations. Analysis confirmed significant effects of Metronome Scaling for both the lag 0, $F(5, 25) = 6.23$, $p = .001$, $\eta_p^2 = .56$, and the lag +1 peaks, $F(5, 25) = 2.58$, $p = .052$, $\eta_p^2 = .34^7$. Generally, these effects reflected a tendency for the most persistent metronome ($\alpha = 1.4$) to produce stronger positive lag 0 peaks and weaker negative lag +1 peaks than the other metronome scaling conditions. More interestingly, unlike the results of Experiment 1,

these functions did suggest the presence of substantial negative correlations at lag +2 (see Figure 12). Analysis confirmed an effect of Metronome Scaling, $F(5, 25) = 7.03$, $p < .0005$, $\eta_p^2 = .58$, on the lag +2 cross-correlations. Here, increases in the metronome scaling yielded increasingly stronger negative correlations between the leaders' current interval and the leader-metronome asynchrony from two taps in the past.

The WDCC functions between the followers' ITI and the follower-leader ASY showed the same difference between synchronization and syncopation observed in Experiment 1, with overall weaker cross-correlations for the syncopation condition (Figure 12). Analysis confirmed a significant effect of Coordination Mode for both the lag 0, $F(1, 5) = 9.78$, $p = .026$, $\eta_p^2 = .66$, and lag +1, $F(1, 5) = 29.49$, $p = .003$, $\eta_p^2 = .86$, with syncopation producing weaker peaks than synchronization. There was also a significant effect of Metronome Scaling on the strength of the lag 0 peaks, $F(6, 30) = 3.51$, $p = .009$, $\eta_p^2 = .41$, with the most persistent metronome condition and the self-paced condition producing stronger peaks than other metronome scaling conditions. More interestingly, the functions also suggested the presence of lag +2 correlations. Analysis of the lag +2 correlations did reveal a significant effect of Coordination Mode, $F(1, 5) = 10.81$, $p = .022$, $\eta_p^2 = .68$, with the synchronization condition producing stronger (negative) lag +2 correlations than the syncopation condition.

In summary, the short-term cross-correlation analyses revealed the same basic patterns observed in Experiment 1. The cross-correlation between the paired interval series revealed the expected peak at lag +1 (or lag 0 for syncopation between follower and leader), and the cross-correlation between the paired interval and asynchrony series showed the expected positive peak at lag 0 and negative peak at lag +1. More importantly, the manipulations of metronome scaling and the coordination mode between participants did affect these patterns of cross-correlation. The more persistent metronome conditions (particularly $\alpha = 1.4$) tended to produce weaker peak correlation between the leaders' tap intervals and the metronome intervals, along with stronger longer-range correlations at lag +2 and lag +3. The more persistent metronome conditions also tended to produce stronger longer-range cross-correlations between the leaders' tap intervals and the leader-metronome asynchrony. In contrast, there were no longer-range cross-correlations for the follower-leader sub-system, and the metronome scaling condition only affected the strength of the peak correlations. As in Experiment 1, the coordination mode also influenced the strength of the cross-correlation between the followers' tap intervals and the follower-leader asynchrony, yielding overall weaker peak correlations for syncopation than for synchronization.

3.3 Conclusion

In contrast to Experiment 1, there were fewer clear symmetries between how the manipulations affected the strength of complexity matching and how they affected the strength of the cross-correlations. Again, the tests of heteroscedasticity suggested significant differences in complexity matching for the follower-leader interaction during synchronization. There was relatively weak complexity matching when the metronome was

⁷This effect was not statistically significant in the adjusted analyses mentioned above ($p = .125$), although the overall pattern of differences in the average peak correlations was the same as in the unadjusted analysis.

anti-persistent or near-random and stronger matching when the metronome was more persistent. In contrast, the strength of the ITI-ITI cross-correlations was near-equally distributed across all levels of metronome scaling, showing only significantly *lower* peak correlation for the most persistent metronome and self-paced conditions. A series of follow-up regression analyses confirmed that, on average, that none of the peak correlations from the cross-correlation analyses significantly predicted the degree of complexity matching (i.e., absolute residual variation scores). This finding stands in contrast to the earlier work on complexity matching in syncopated tapping (Coey et al., 2016), which did find that peak cross-correlations predicted the strength of complexity matching. This difference is possibly related to the overall level of complexity matching observed, as even in the earlier study, the peak correlation was only a significant predictor of complexity matching when that matching was relatively weak ($R^2 \approx .40$).

There was, however, some interesting relationships between the cross-correlations and the asynchrony scaling. As in Experiment 1, the syncopation condition produced both more persistent asynchrony and weaker patterns of interval-asynchrony cross-correlation for the follower-leader sub-system. These findings might be taken to mean that when the follower is relatively *less* coupled to the leader, their coordination is relatively more persistent. To the contrary, follow-up tests showed that there were substantial correlations between the asynchrony scaling and peak cross-correlations (mean $r = -.44$), indicating stronger (i.e., more negative) lag +1 peaks were associated with more persistent asynchrony series, within the syncopation condition. In contrast, there were generally weak correlations between the asynchrony scaling and the peak cross-correlations for synchronization (mean $r = .11$). As in Experiment 1, these findings demonstrated that even though certain manipulations (e.g., coordination mode) might not have pronounced effects on complexity matching in the interval series, they might still influence the asynchrony dynamics.

4. General Discussion

The purpose of the current study was to address two limitations in the past research on complexity matching. Namely, this study was designed to explore the complexity matching effect outside of synchronization tasks and examine the fractal scaling of the asynchrony series. The results of these experiments were generally consistent with the findings of the past research, but also provided new insights about potential upper limits on the degree to complexity matching, the possible modes of coupling that might underlie complexity matching, and the relationship between the structural dynamics of interval and asynchrony behavioral time-series. Below, we briefly summarize the key findings with regards to previous research and then discuss each of these new contributions to the complexity matching literature in turn.

The manipulations of the magnitude of variation and fractal scaling of the metronome did affect the complexity matching between the interval series, but generally did so in a similar way for synchronization and syncopation. This finding is consistent with the one previous study exploring complexity matching in syncopated tapping (Coey et al., 2016). This supports the conclusion that complexity matching is not specific to synchronization tasks, in which participants are explicitly trying to match their behavior to the timing signal exactly.

During the syncopated mode of coordination, the participant must generally adjust their behavior to the dynamics of the timing stimulus, but the exact time of their taps is not directly specified by a perceptible stimulus event (Coey et al., 2016). So, complexity matching seems to be a more general consequence of coordination between complex systems, occurring even when the coordination is likely to be weaker (i.e., during syncopation). Also in line with past research (Coey et al., 2014; Marmelat et al., 2014), there was weaker matching when the metronome presented an anti-persistent and random signal, and stronger matching when the metronome was more persistent. Moreover, when the follower was synchronized, complexity matching was strongest when the leader produced tapping intervals near the level of scaling associated with self-paced tapping ($\alpha \approx 0.7$).

These findings might be taken to support the conclusion that complexity matching depends on the “intrinsic scaling” of the systems (Kello & Van Orden, 2009). It is important to note, however, that several studies have shown that timing behaviors not only display (mono)fractal scaling, but also a more complex *multifractal* structure (Kelty-Stephen, Palatinus, Saltzman, & Dixon, 2013; Stephen & Dixon, 2011). In short, multifractal dynamics are evidence that these performances are not defined by a one single value of fractal scaling, but by a spectrum of scaling values, which might challenge the notion of a truly “intrinsic” value of fractal scaling. This is particularly important because the matching of the multifractal structure can be taken to mean that the two systems are actually coupled at many timescales of fluctuation in their behavior, i.e., global coordination (Stephen & Dixon, 2011). Thus, truly understanding the possibility of a “preferential” value of scaling in the timing stimulus for maximizing the strength of complexity matching must involve future research considering how such experimental manipulations also influence multifractal dynamics.

As mentioned above, the present results indicate that there are certain limits on the participants’ tapping behavior and the degree to which complexity matching can occur for strongly persistent signals. Experiment 1 demonstrated a lower limit in the variability of the tapping intervals ($\approx 5\%$ *CoV*) and, moreover, that the participants tended to be slightly more variable than the metronome across all conditions. Experiment 2 revealed an upper limit in the scaling of the tapping intervals ($\alpha \approx 1$) and, moreover, that the participants tended to be more random than the metronome across all conditions (also see Washburn et al., 2015). One possible explanation of these limits is that the behavior of the response system mirrors the complexity of the timing system, along with a small amount of additional random variation due to “errors” in the coordination. Past research has demonstrated that injecting random variation into a task will result in more random behavior (e.g., Kello, Beltz, Holden, & Van Orden, 2007), but this assertion seems to be at odds with the other finding that the coordination (i.e., the asynchrony) is not random error, but itself a persistent, fractal signal (also see Chen et al., 1997).

One approach to understanding these results is to explore models of fractal scaling in timing behaviors that have not yet been applied to the study of complexity matching. Delignières and Torre (2008) have demonstrated that modifying older discrete timing models (Vorberg & Schulze, 2002; Wing & Kristofferson, 1973) so that the centralized clock component contains a fractal signal can account for both the fractal scaling in the intervals during self-

paced tapping and in the asynchrony during synchronization tapping. Delignières, Torre, and Lemoine (2009) showed that similar models can also account for finding increased persistent scaling in the asynchrony during syncopation tapping (Chen et al., 2001). These models, and others based on a short-term error correction process (e.g., Hennig, 2014), could potentially be applied to understanding the current results, which might, in turn, inform the future development of these models. For instance, one interesting question would be whether the local adjustments would explain the partial complexity matching in the longer-timescale fluctuations at lower magnitudes of variation in the timing stimulus observed in Experiment 1.

It is also important to explore whether these limits on the emergence and strength of complexity matching behavior reflect inherent stabilities of a dynamical system. To this end, further development of emergent, dynamical models that capture these stabilities and account for the complexity matching results (Delignières, Torre, & Lemoine, 2008; Repp & Steinman, 2010; Torre & Balasubramaniam, 2009), is also an important pursuit. For instance, it is also possible that the partial complexity matching observed at low magnitudes reflects a perceptual sensitivity (Marmelat et al., 2014) to longer-range fluctuations in the timing stimulus. This type of coupling across many timescales is the central thesis of the dynamical models of “global” coordination that do not rely on localizable, mechanistic coordination processes (Stephen & Dixon, 2011). These insights might ultimately speak to the conceptualization of “local” and “global” coordinative processes.

Currently, in the complexity matching literature, the term “local coordination” is used in conjunction with short-range cross-correlation between the interval series (e.g., Marmelat & Delignières, 2012), or, less often, in conjunction with instantaneous relative phasing between the two movement series (Fine et al., 2015). The term “global coordination” is often used in conjunction with the correlation between the two fractal scaling exponents for the interval series, which are themselves defined across the entire trial. The correlation between the (mono)fractal scaling exponents, however, does not directly address the deeper contention of the theory of global coordination that complexity matching ultimately reflects a *coupling* of two systems across many different scales. As mentioned above, evaluating this hypothesis might be better done by examining the multifractal dynamics of the timing behaviors (Stephen & Dixon, 2011). What is less clear, however, is the assertion that because the behaviors are correlated at across many scales, the two systems must be also *coupled* at these all of these scales. Stephen and Dixon (2011), who strongly argue for global coordination, acknowledge that the coupling between systems might be “local” (without necessarily also being the result of a representative, mechanistic process), while the coordination between systems was “global”, and that the two might entail one another. This means that, aside from deeper debates between the mechanistic and nomothetic perspectives (see Diniz et al., 2011), it is essential to better understanding complexity matching to determine the timescales at which the perceptual information operates.

From the current experiments, it seems that, at one scale, participants might be attuned to the longer-range variation in the metronome intervals, which constitute the coherent oscillatory patterns salient in the metronome beats. At another scale, participants seem sensitive to their immediate asynchrony to the metronome. Neither one of these timescales must take

theoretical priority. Instead, a more informative approach would be to clearly define how the participants are perceptually coupled at these different timescales, and how these timescales of coordination contribute to governing the tap performances.

Finally, the present results raise an important question concerning the differences in the asynchrony scaling, and its relation to the complexity matching of the intervals. Specifically, both Experiments 1 and 2 demonstrated that the scaling of the follower-leader asynchrony differed substantially between the synchronization and syncopation condition, while there was little to no difference in the overall strength of the complexity matching of the intervals. Similarly, Experiment 1 showed that the leader-metronome asynchrony scaling decreased in a gradual, linear fashion with increases in the magnitude of variation in the metronome, while the complexity matching in the intervals emerged abruptly at a critical level of magnitude. Again, these results show that complexity matching can be equally evident in conditions that produce substantial differences in the coordination (i.e., asynchrony) dynamics. It is important to note, however, there were some more subtle relationships between the coordination mode and complexity matching. For instance, the results of Experiment 2 showed that only follower-leader synchronization lead to significant heteroscedasticity in the complexity matching, and only syncopation involved a significant relationship between the strength of the short-term crosscorrelations and the strength of complexity matching. Taken together, these findings therefore raise a critical issue for future research; how to model the interval *and* asynchrony series as a set of interrelated dynamical variables in order to understand the nature of the coordination underlying complexity matching.

One potentially fruitful approach would be to consider these time series within the framework of behavioral dynamics models (Warren, 2006). In brief, behavioral dynamics asserts that coupled systems can be understood at multiple levels of description. At one level, the two systems can be considered as being separate, but engaged in a relatively high-dimensional interaction with one another. At another level the two systems can be understood as one unified system, whose behavior display relatively low-dimensional dynamics. Critically, the low-dimension behavioral dynamics are thought both to emerge from and constrain the high-dimensional interaction in a form of reciprocal causality. Here, it might be appropriate to conceive of the interval series, and their cross-correlation and complexity matching, as describing the high-dimensional interaction of two inherently complex systems. The asynchrony series might serve to describe the low-dimensional dynamics capturing the behavior of the larger, unified system (Chen et al., 1997). A goal for future research from this perspective would be to understand how the asynchrony dynamics might emerge from the high-dimensional interaction of the coupled systems, and how that stable coordination in turn constrains the behaviors of the individual systems, ultimately leading to complexity matching.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Appendix

Appendix A

The matching algorithms were implemented to ensure unique pairings of observations in a timing signal series (e.g., metronome beat onset times) with observations in a response series (e.g., participant tap onset times). The need for matching was due to occasional deviations from the intended 1:1 ratio of recorded taps to timing events, when a participant lost attention and did not tap to a particular beat or forgot to release the key fully. Separate algorithms were devised for the synchronization and syncopation conditions.

Synchronization

The algorithm started by working through observations in the timing series sequentially. For each timing observation, the absolute time differences with all responses were calculated. The response with the smallest absolute time difference was considered the candidate matching response for that timing observation.

It was possible that there were two candidate matching responses for a particular timing observation. This would occur if there were two responses with equal absolute time differences (i.e., two responses occurring the same period before and after the current timing observation). This did not occur often, but to address such cases, the algorithm would implement a rule for choosing the preferred candidate. This rule involved considering the previous and next timing observations, which was important due to the fact that there was variation in the timing intervals, as well as in the responses. This process is depicted in Figure 1A. When one candidate was closer to the current timing observation than to the next closest timing observation (left column), that candidate was selected. More ambiguous situations were also possible, however, as when both candidates either closer to the next closest timing observation than to the current timing observation (top right) or both closer to the current timing observation than to their next closest timing observation (bottom right). In all these ambiguous cases, the later candidate was selected.

Next, the algorithm worked with the array of candidate timing-responses matches. All of the values in the timing series were unique, but it was possible that some would share the same candidate response match. In this case, the match pair with the smallest asynchrony was selected, and the others were excluded. The result was final series of unique timing-response matches, which were used to calculate the interval and asynchrony series submitted to further analysis.

Syncopation

This algorithm relied on alternation between timing and response onset times, which is a primary characteristic of the syncopation task. This algorithm first combined the onset times

from the timing and response series into a single array (along with a marker indicating to which series the observation belonged), and then sorted by onset time so that the combined array would ideally alternate between timing and response observations. Then, working forward sequentially from the first timing series observation, the algorithm checked whether a response observation fell, at any point, between the current and next timing observations. If not, the current timing observation was excluded, and the process advanced to the next timing observation. If more than one response fell between two timing observations, the earlier of the two was selected to be matched to the current timing observation. This produced a set of matched timing-response pairs, with the response always falling after the timing event. The combined vector was then divided back into separate timing and response series, and the interval and asynchrony series were calculated for further analysis.

Manipulation Check

Finally, we wanted to provide a simple test assessing any possible effects of the matching procedure on the results of later analyses. In particular, we were concerned with how matching might alter the estimates of the fractal scaling exponents used to assess complexity matching. So, we estimated fractal scaling exponents from the unmatched (but otherwise pre-processed) interval time series, and calculated simple regression models predicting the matched scaling exponents from the unmatched scaling exponents. These tests revealed very strong linear relationships between the matched and unmatched scaling exponents for both the interval series from Experiment 1 ($R^2 = .96$) and those from Experiment 2 ($R^2 = .96$). The scatterplots portraying these relationships are presented in Figure 2A. These results suggest that while matching might have some small effect on estimating the fractal scaling of the interval series, both the matched and unmatched series would reveal the same general pattern of results in subsequent analyses.

Appendix

Appendix B

Detrended fluctuation analysis (DFA) is one common technique for evaluating fractal, power-law scaling in a time-series of behavioral performances. It is important to note that DFA often necessitates “integration” (i.e., cumulatively summing) the time-series prior to analysis. This is necessary because DFA works on the diffusive property of fractional Brownian motions. Thus, time series like the inter-tap intervals involved here (i.e., fractional Gaussian noises) need to be integrated for DFA to provide accurate estimates of the fractal scaling.

DFA is a time-domain technique, and uses windows of different numbers of observations to examine the timescale of variation in the series. The series is first broken into windows of some minimal size, and within each window a local linear trend is computed, and the root mean square variation around this linear trend is calculated. The root mean square variation is then averaged across all windows to yield a summary score, referred to as “F”. Then the window size is increased and the process is repeated, iteratively, up to windows of some maximum size. These “F” (average residual variation) scores are then plotted against the window sizes on a double-logarithmic DFA plot, with large window, slow timescales

represented on the right and small window, fast timescales represented on the left side. Here, the DFA plot generally only showed a single linear scaling region across all window sizes (Figure 1B), although it is not uncommon for the plot to show distinct sub-regions with different linear slopes. The slope of the regression line fitting the DFA plot is taken as an estimate of the Hurst exponent defining the power-law relationship (Eq. 1) between the variation in the time series (x) and the timescale over which the series is observed (Δt), with ideal fractal scaling defined by $H = 1$.

$$\text{Var } x(t) \propto \Delta t^{2H} \quad (\text{Eq. 1})$$

The Hurst exponents output by the DFA algorithm is often rescaled to be directly comparable to the scaling exponent (α) with a simple linear transformation ($\alpha = 2*(H - 0.5)$), so that random, “white noise” variation in the time series is indicated by $\alpha \approx 0$, persistent, “pink noise” variation is indicated by $\alpha \approx 1$, and anti-persistent, “blue noise” variation is indicated by $\alpha \approx -1$.

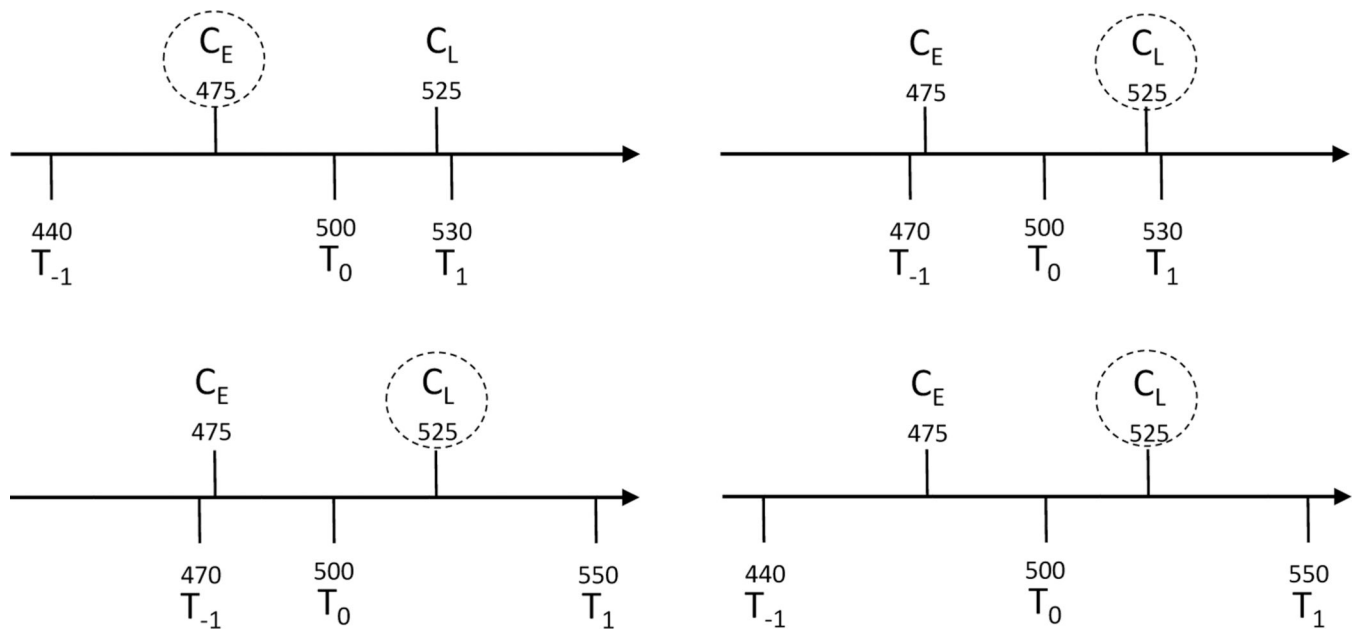
In the present experiments, we chose 16 observations as the minimal window size and limited the maximal window size to $1/4^{\text{th}}$ the length of the time series (i.e., 128 observations). Examination of the DFA plots generally suggested a single linear scaling region, and so the entire DFA plot was used to estimate the Hurst exponent. Hurst exponents were rescaled into α , and the inferential statistical tests were performed on these α scaling exponents to assess how the experimental manipulations affected the fractal scaling of the finger tapping time series.

Windowed detrended cross-correlation (WDCC) is a modified version of a standard cross-correlation technique (XCOR), which assess the short-term dependencies between two time series of behavioral performances. Both of these techniques assess the statistical relations between two time series as a function of the time lag between observations. As with a standard (Pearson) correlation, the cross-correlation coefficient captures the both direction and strength of the relationship between the two series, ranging from -1 to 1 . Cross-correlation at lag 0 captures the relation between the sequentially matched observations. Cross-correlation at positive lags captures the relation between the current observation in some referent series (e.g., the metronome) and later observations in some comparison series (e.g., the leader participant). For instance, a positive cross-correlation at lag +1 indicates that increases in the current metronome inter-onset interval are related to increases in the next leader inter-tap interval (i.e., the metronome leads the participant by one sample). Conversely, cross-correlation at negative lags captures the relation between later observations in the referent series and the current observations in the comparison series.

WDCC analysis attenuates the strength of the cross-correlation between the two series that might be due to linear trends in the data and long runs of observations. Both time series are broken into small windows (e.g., 15 observations) and a local linear trend is removed from each window. The cross-correlation procedure is then performed within each window, and the cross-correlation function (i.e., the pattern of coefficients across all lags) is averaged across all the windows. The resulting cross-correlation function is therefore not influenced

by trends and long runs of observations, which essentially highlights only the short-term correlations between the two series and often eliminates longer-term correlations (Figure 2B).

In the present experiments, we performed WDCC analysis on the sets of the inter-event intervals series (e.g., metronome inter-onset intervals and leader inter-tap intervals) as well as on the sets of asynchrony and “response” inter-event interval series (e.g., metronome-leader asynchrony and leader inter-tap intervals) in order to assess all of the statistical dependencies between the coupled systems. In all cases, the maximum time lag was ± 10 observations and the window size was 15 observations. The windows were non-overlapping, and a local linear trend (i.e., least-squares regression) was removed from each window prior to cross-correlation. When appropriate, the coefficients at specific time lags were submitted to inferential statistical tests to assess how the experimental manipulations affected the short-term relationships between the coupled systems.

**Figure 1A.**

Four possible versions of a multiple match candidate scenario, indicating hypothetical onset times for the two candidate response matches (earlier, C_E and later, C_L), their equidistant relationship to the current timing observation (T_0), as well as the previous (T_{-1}) and next (T_1) timing observations. The dashed circles represent the selected candidate response.

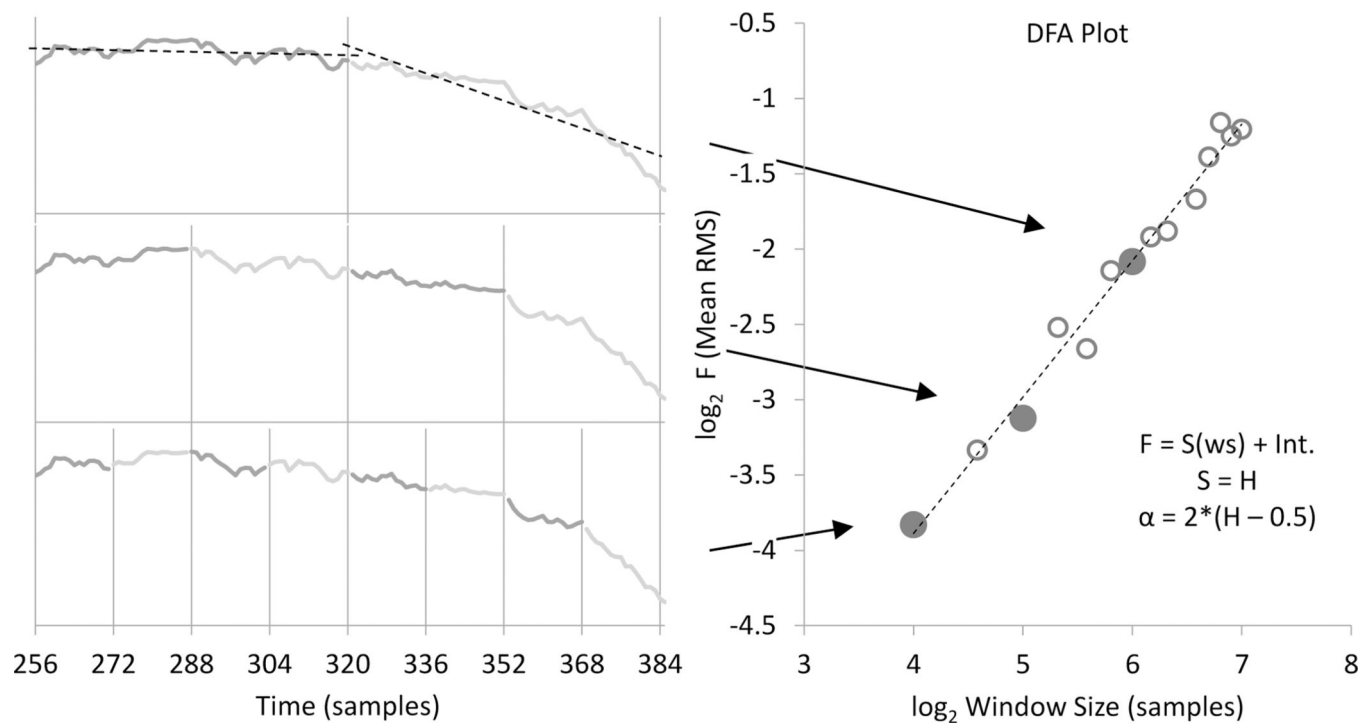
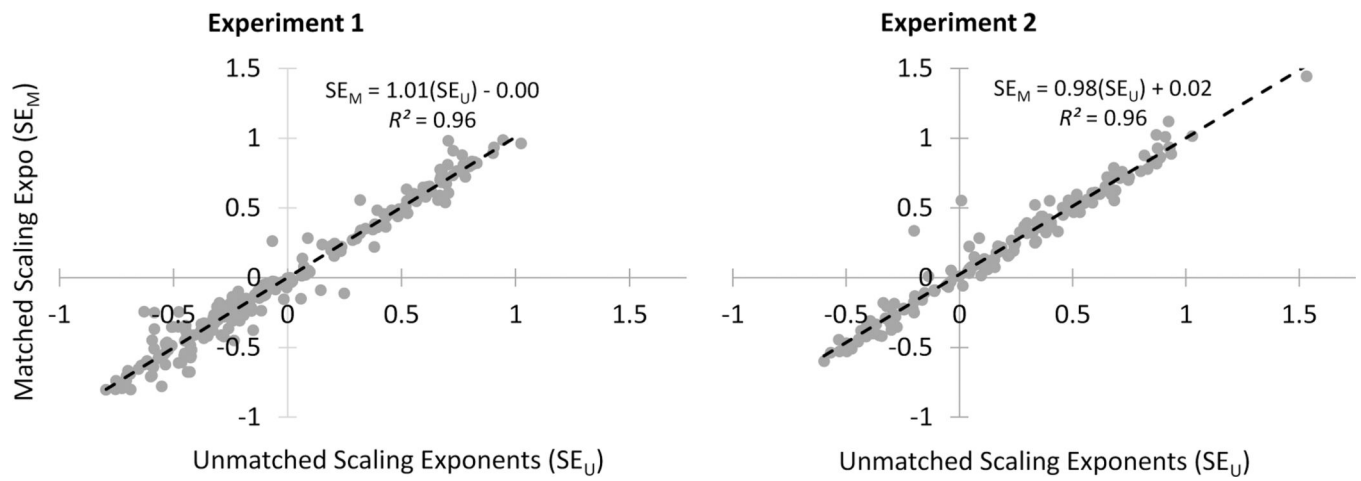


Figure 1B.

An example of detrended fluctuation analysis showing a time series broken into three different window sizes (16, 32, and 64 points), with local linear trends displayed for the largest window size, and the associated scaling plot.

**Figure 2A.**

Scatterplot of the relationship between fractal scaling exponents generated from the unmatched and matched interval series from Experiment 1.

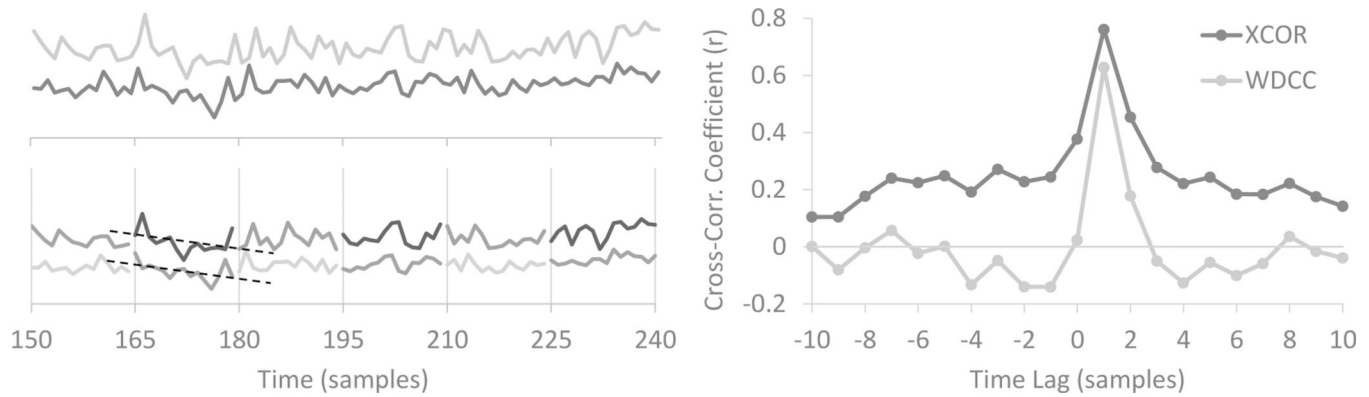


Figure 2B.

A short period of two empirical ITI series (top left), the same series divided into small windows with local trends displayed within the second window (bottom left), and the associated XCOR and WDCC functions across lags of -10 to $+10$.

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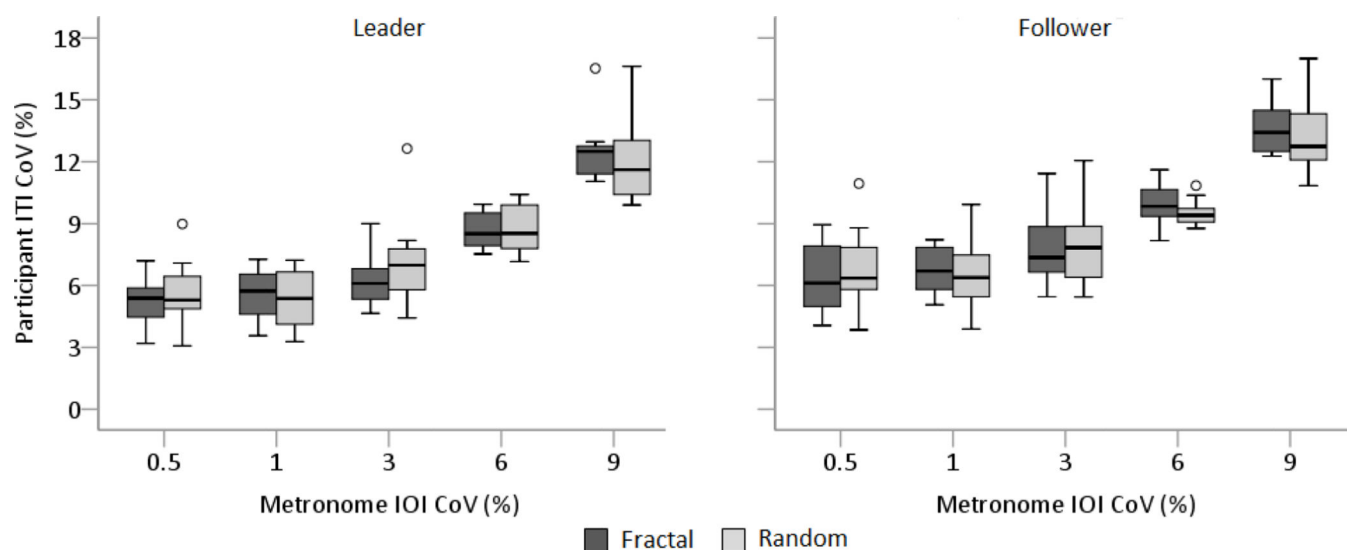


Figure 1.
The coefficient of variation for the leader and follower ITI series as a function of Magnitude and Metronome Scaling (FL sub-system) in Experiment 1.

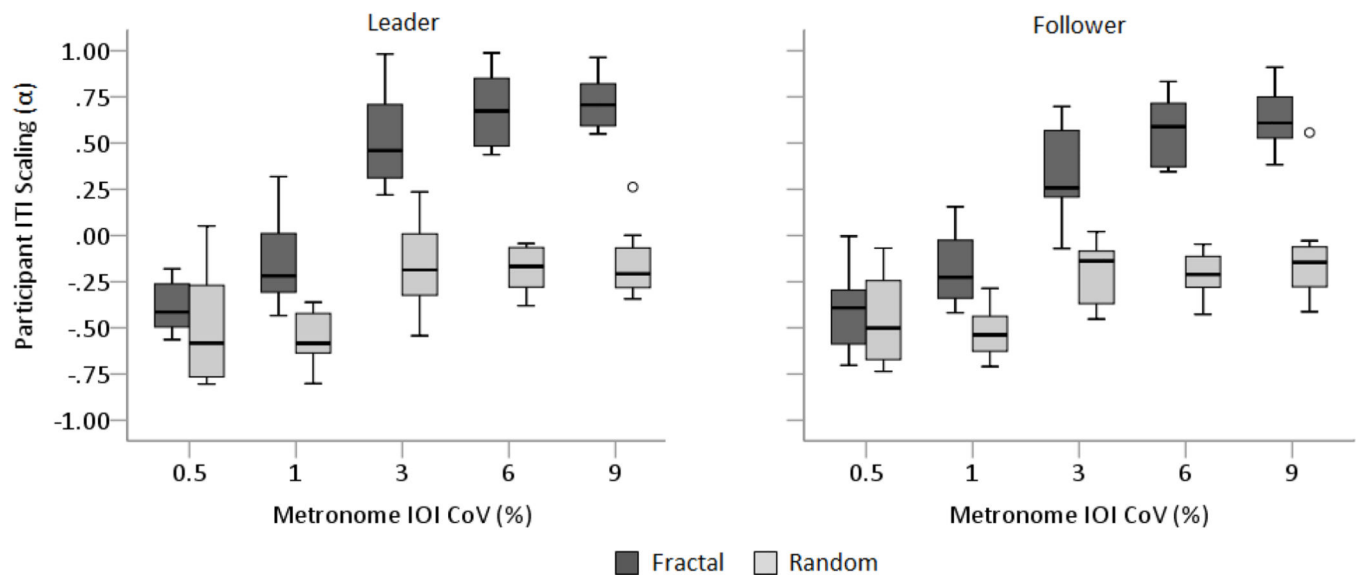


Figure 2.

The fractal scaling of the leader ITI (left) and follower ITI (right) series as a function of Magnitude and Metronome Scaling (FL sub-system) in Experiment 1.

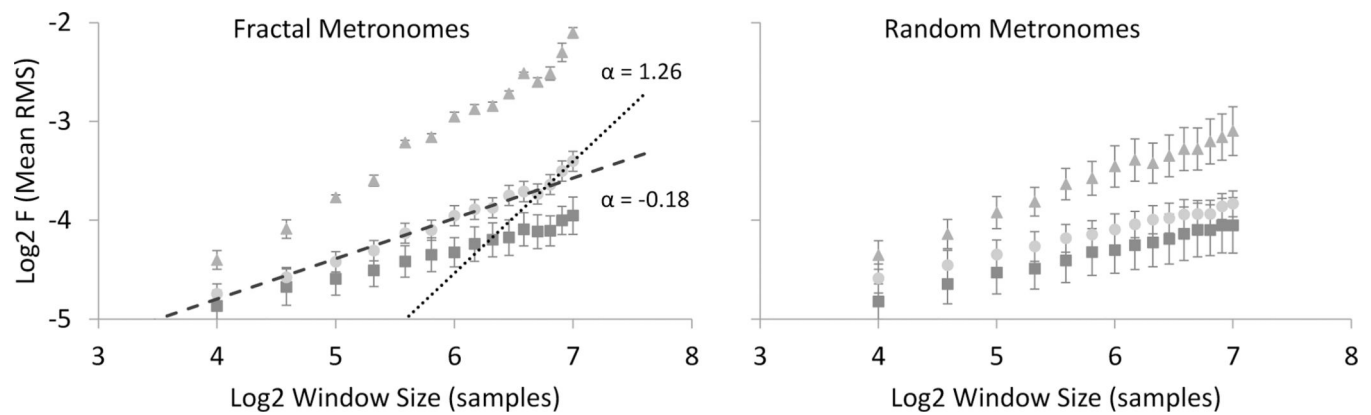
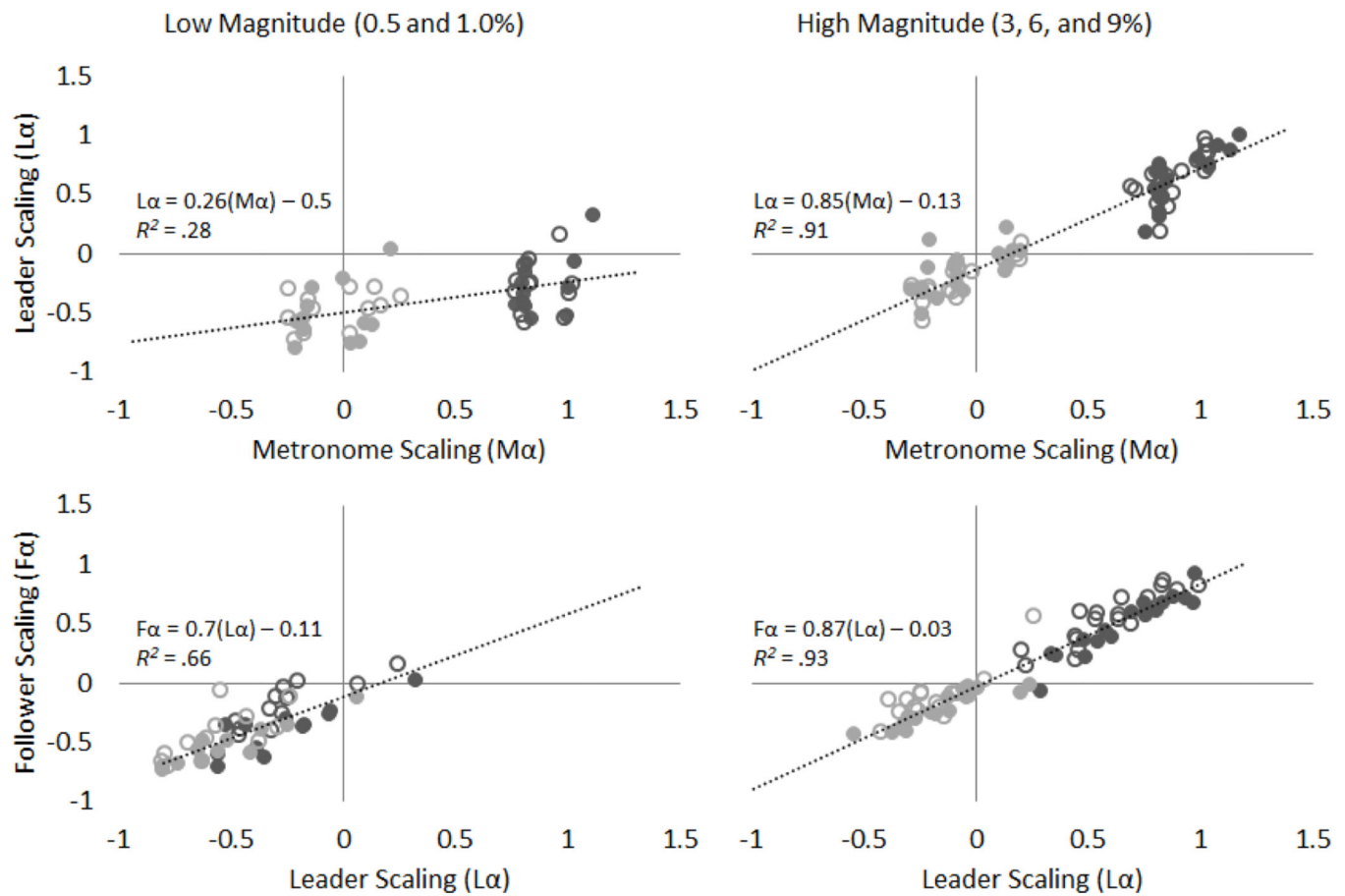


Figure 3. Average scaling plots from detrended fluctuation analysis of the leaders' inter-tap interval series from the fractal (left column) and random metronome conditions (right column) as a function of Magnitude in Experiment 1.

**Figure 4.**

Complexity matching scatterplots for both the leader-metronome (top row) and the follower-leader sub-system (bottom row) in Experiment 1, with the fractal metronomes represented in dark grey and the random metronomes represented in light grey, and with synchronization represented in filled-in circles and syncopation represented in empty circles.

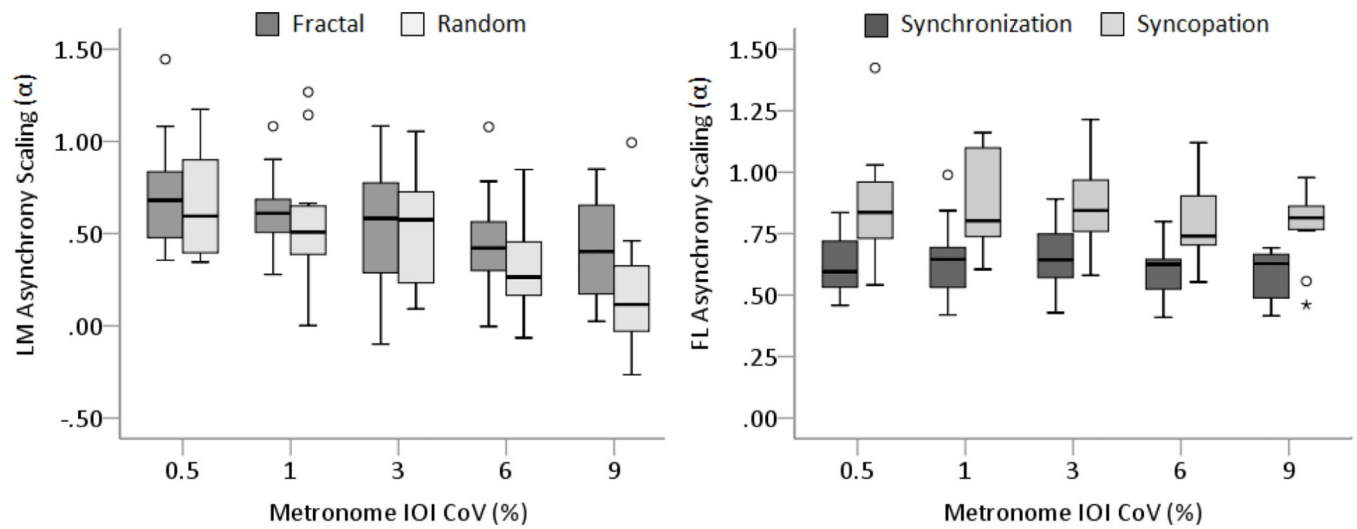


Figure 5.

The scaling exponents (α) for the leader-metronome asynchrony series as a function of Magnitude and Metronome Scaling (left panel) and for the follower-leader asynchrony series as a function of Magnitude and Coordination Mode (right panel) in Experiment 1.

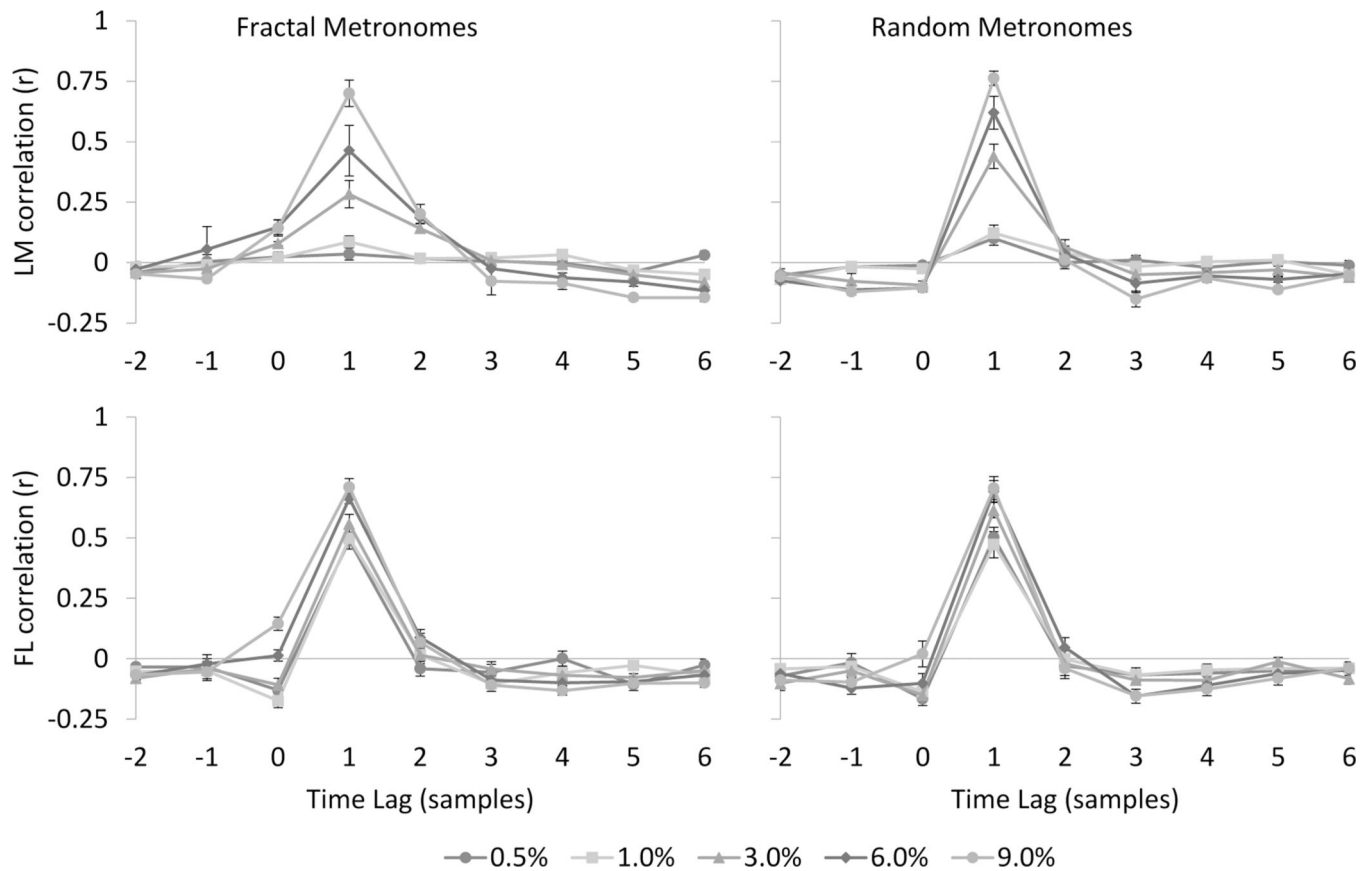


Figure 6.

Windowed detrended cross-correlation between the paired interval series for the leader-metronome (top row) and follower-leader (bottom-row) sub-systems for both the fractal (left column) and random (right column) metronome conditions in Experiment 1. These plots represent the synchronization condition only. The syncopation condition produced nearly identical functions, only shifted right one time lag for the follower-leader sub-system.

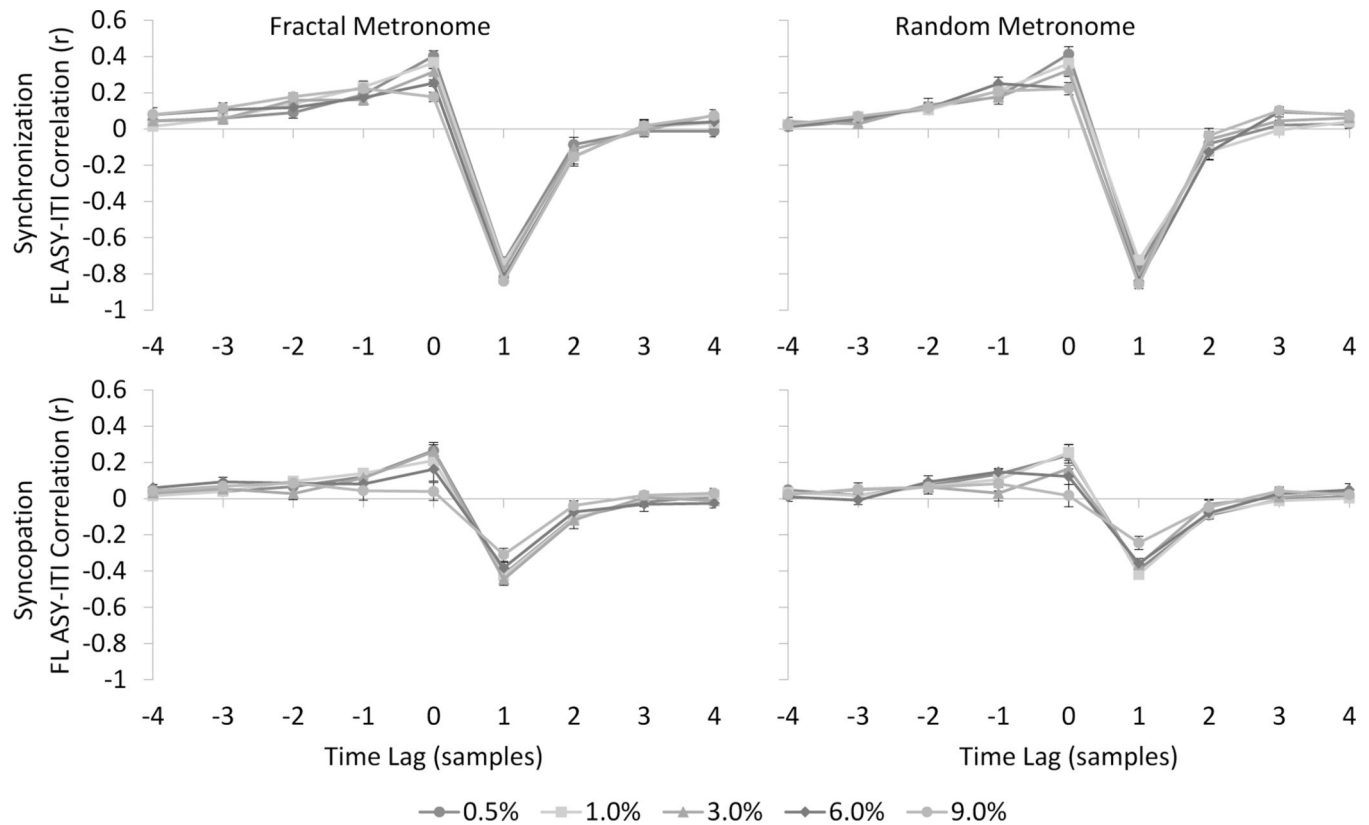


Figure 7.

Windowed detrended cross-correlation between the paired asynchrony and interval series for the synchronization (top row) and syncopation (bottom-row) conditions for both the fractal (left column) and random (right column) metronome conditions in Experiment 1. These plots represent the follower-leader sub-system only. The leader-metronome sub-system produced nearly identical functions to those of the synchronization condition portrayed here.

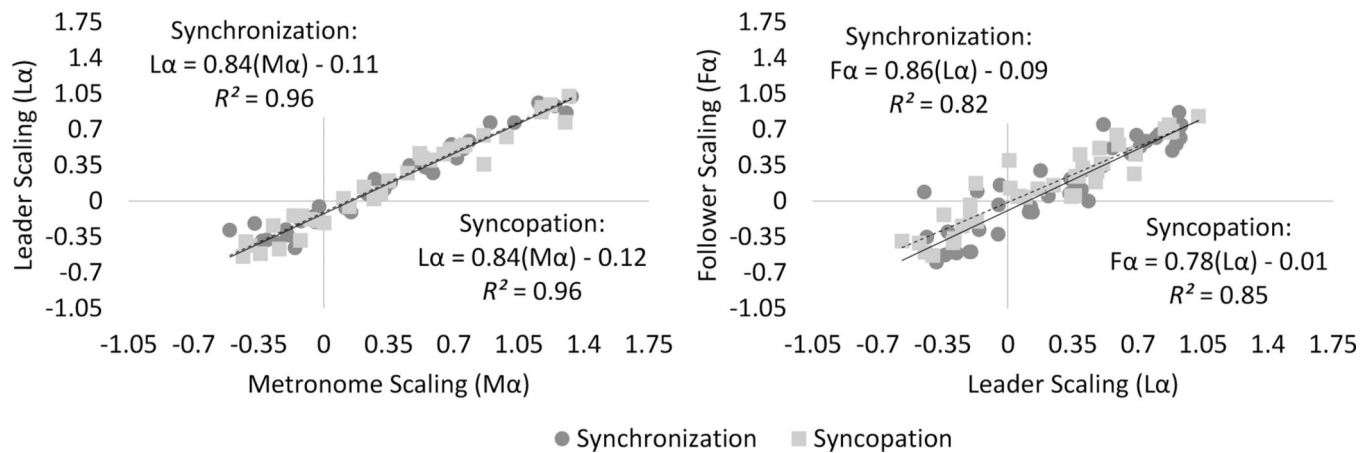


Figure 8.

Complexity matching scatterplots for the leader-metronome (left) and follower-leader (right) sub-systems for both the synchronization (dark grey circles) and the syncopation (light grey squares) conditions in Experiment 2. The regression lines are solid for synchronization and dashed for syncopation.

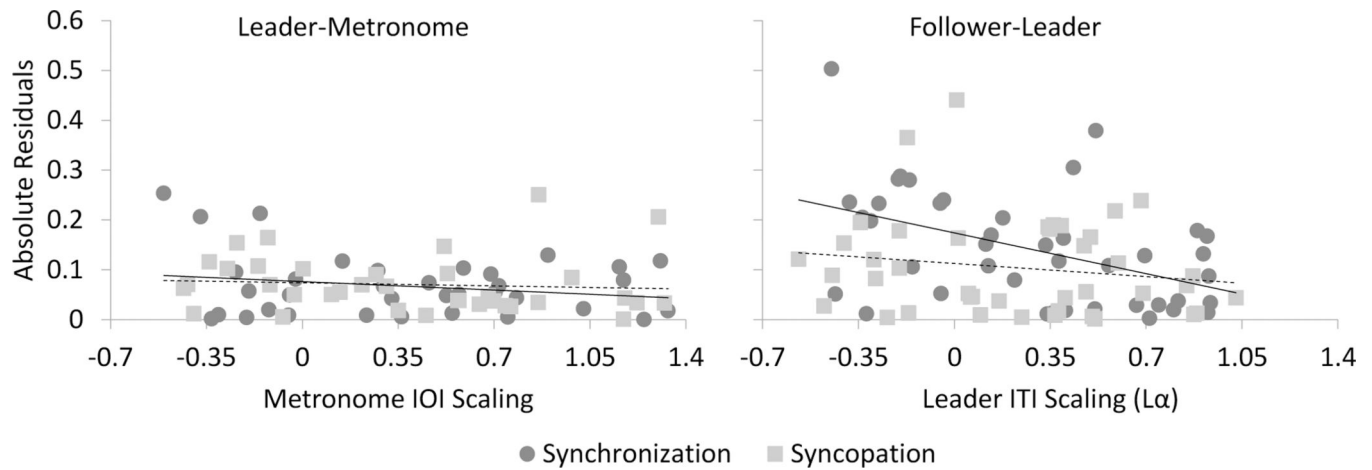


Figure 9. Scatterplots of the absolute residual variance around the complexity matching regression line as a function of the scaling in the referent inter-event interval series for the leader-metronome (left) and follower-leader (right) sub-systems in Experiment 2. The synchronization condition is represented in dark grey circles (solid lines) and the syncopation condition is represented in light grey squares (dashed lines).

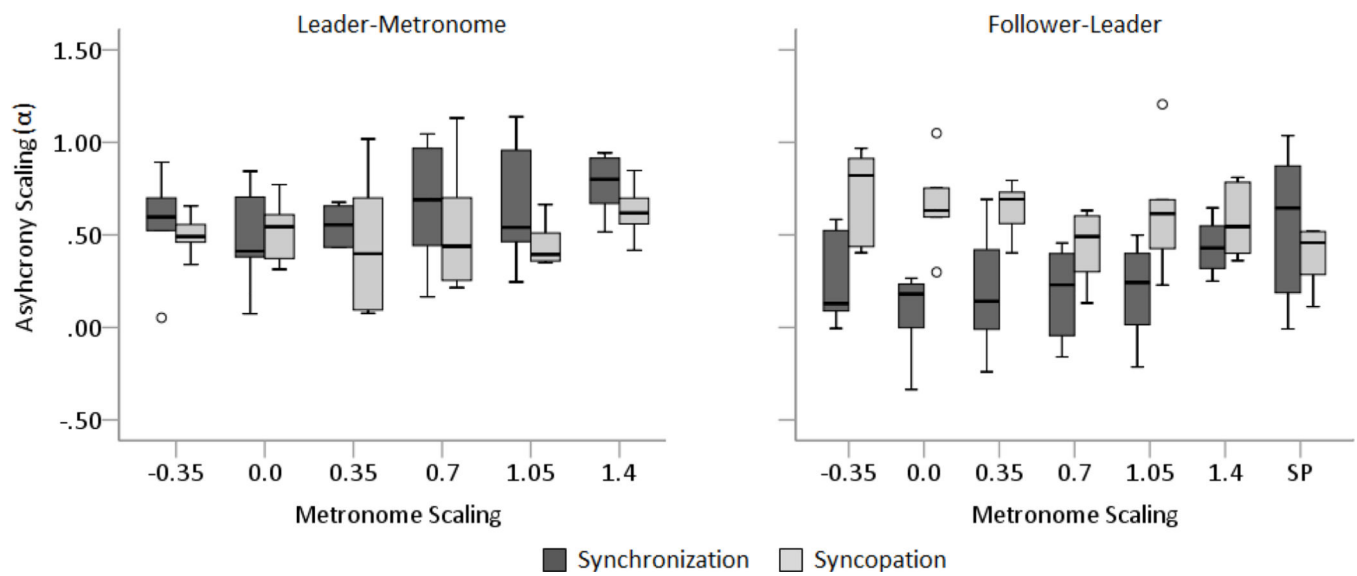


Figure 10.

The scaling exponents (α) asynchrony series for the leader-metronome (left panel) and the follower-leader (right panel) sub-systems as a function of Metronome Scaling and Coordination Mode in Experiment 2.

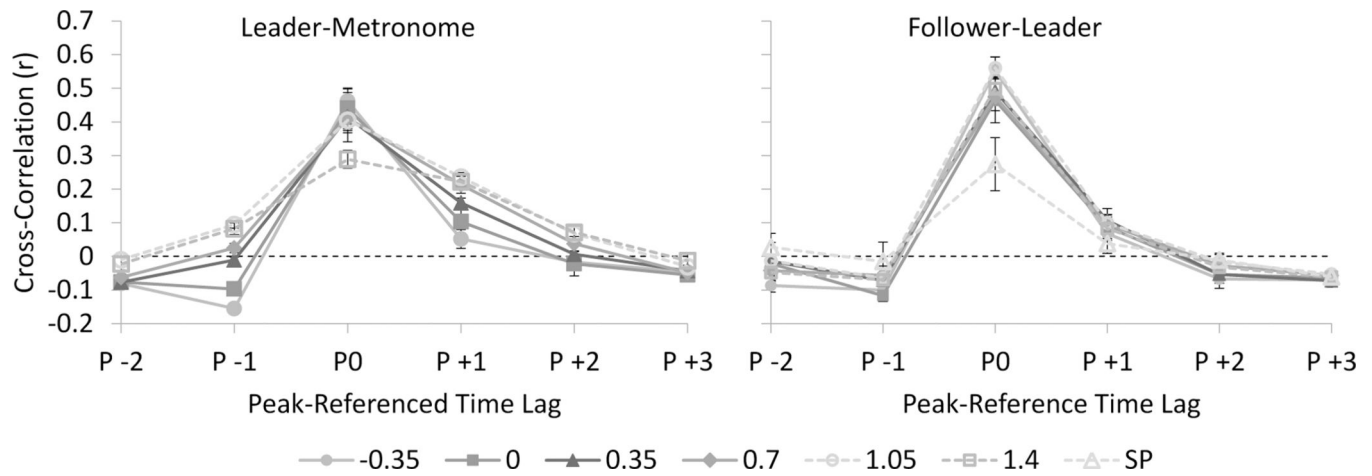


Figure 11. Windowed detrended cross-correlation between the paired interval series for the leader-metronome (left) and follower-leader (right) sub-systems as a function of Metronome Scaling in Experiment 2.

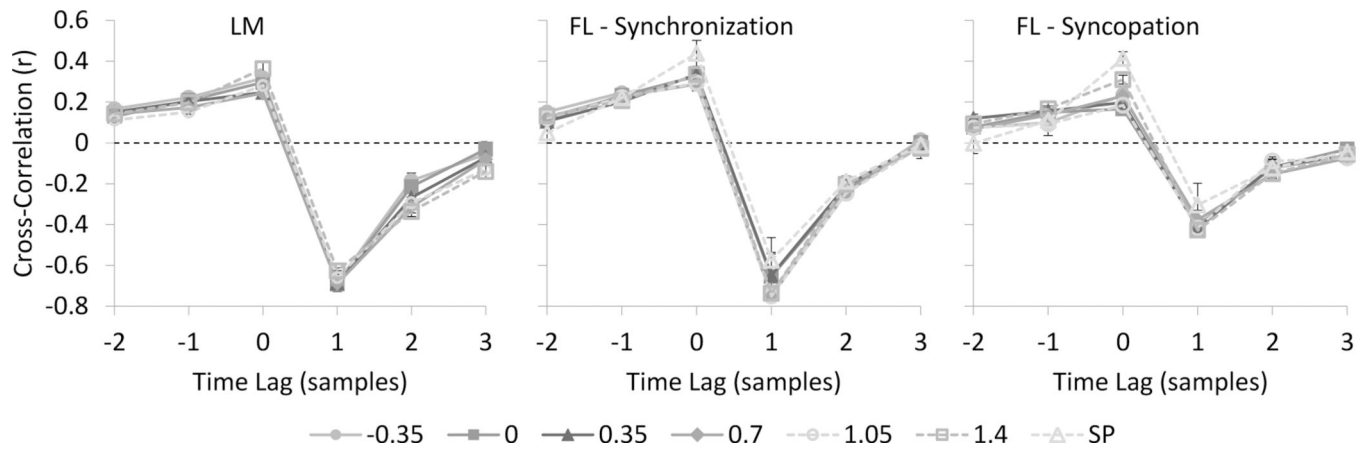


Figure 12.

Windowed detrended cross-correlation between the paired asynchrony and interval series, as a function of Metronome Scaling, for the leader-metronome sub-system (left) and the follower-leader sub-system during synchronization (center) and syncopation (right) in Experiment 2.